

Modulação e Valor da Energia

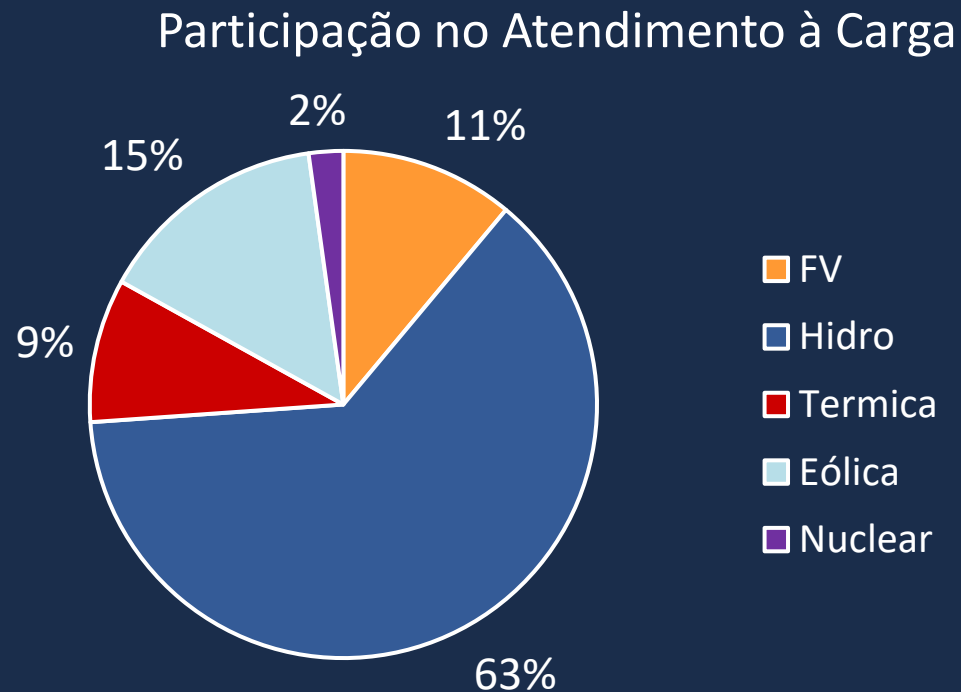
Aplicação da Metodologia de Modulação para
avaliar benefícios de diversas fontes para a
flexibilidade no SIN

Autor: Jonathan Veiga Velasco

9/06/2025

Matriz Elétrica

- Período de estudo desde 01/2024 a 05/2025
- Carga média no Sistema Interligado Nacional foi de 80 052 MWm



Serviço de Modulação

- Mede o benefício da participação de cada fonte.
- Fator de participação de cada fonte:

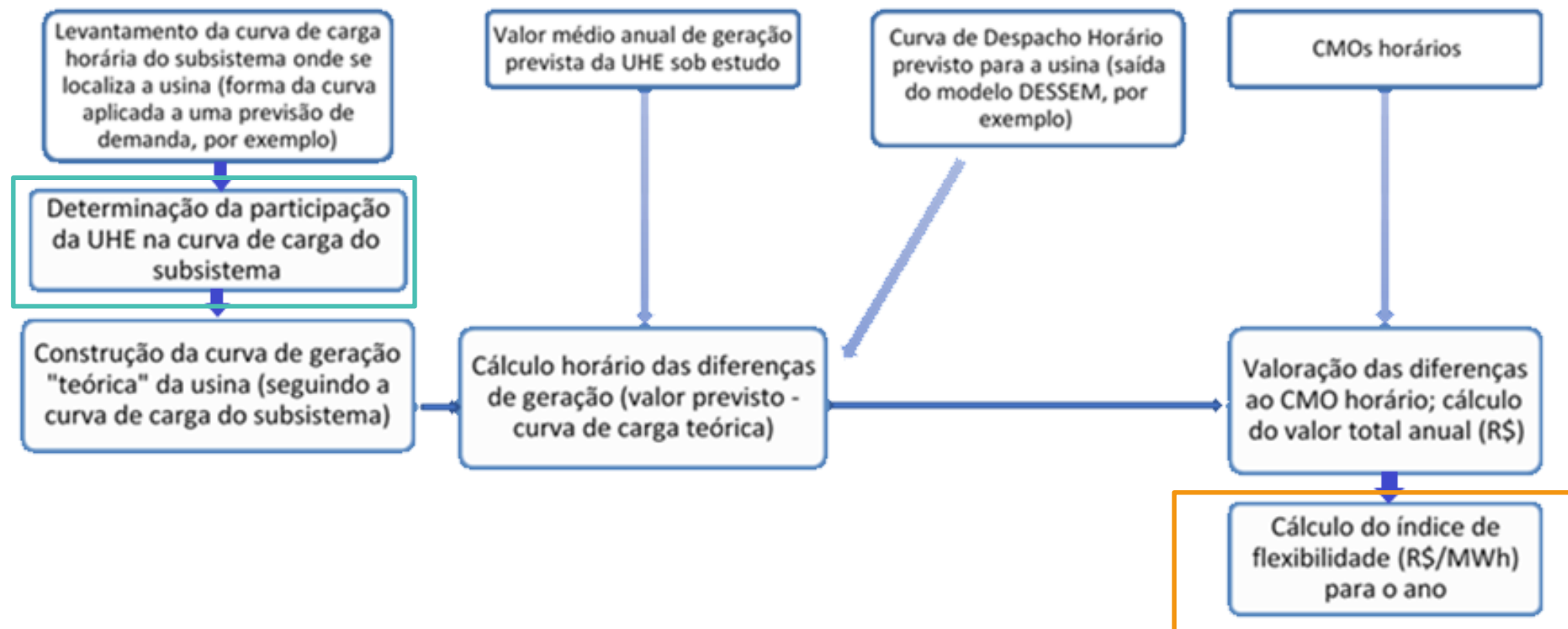
$$k_{part} = \frac{\sum E_{ger}}{\sum E_{dem}}$$

- $V_{Mod} = p_{CMO} \cdot (k_{part} \cdot E_{dem} - E_{ger})$

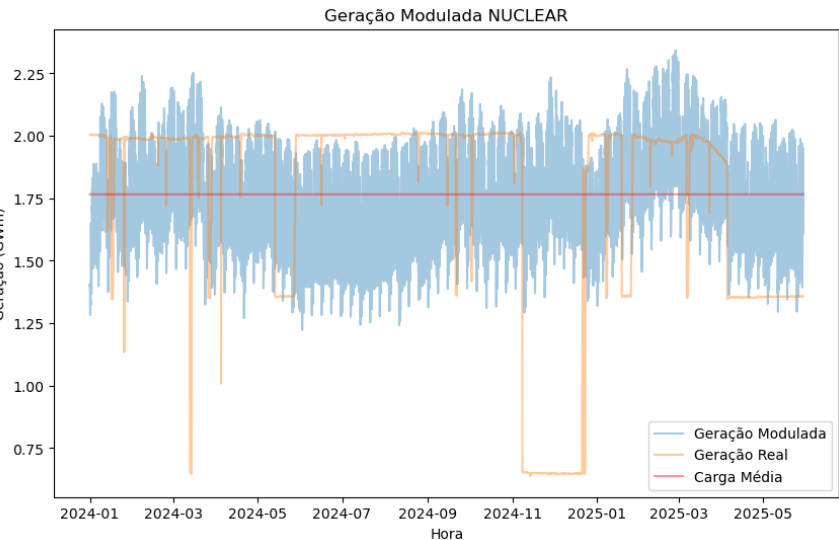
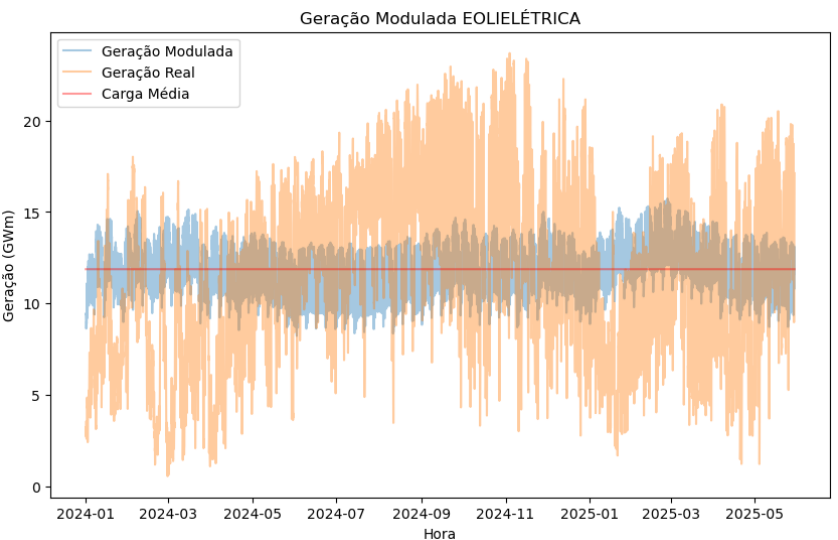
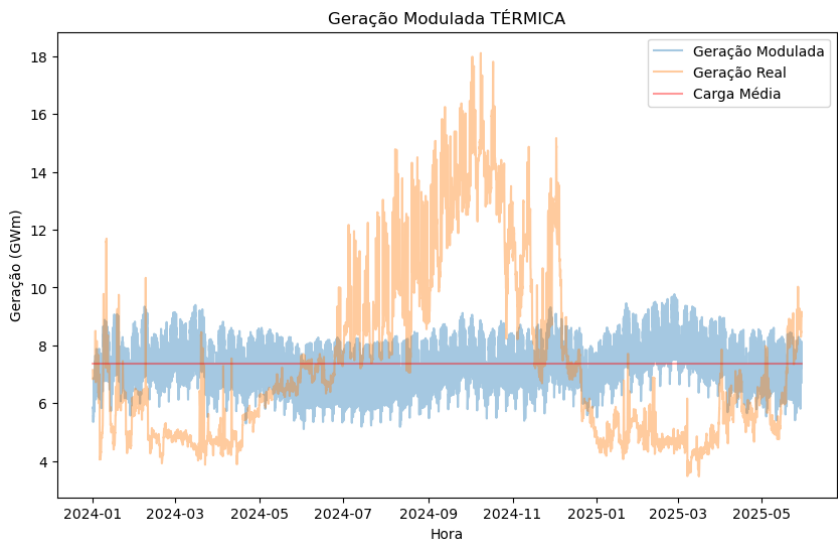
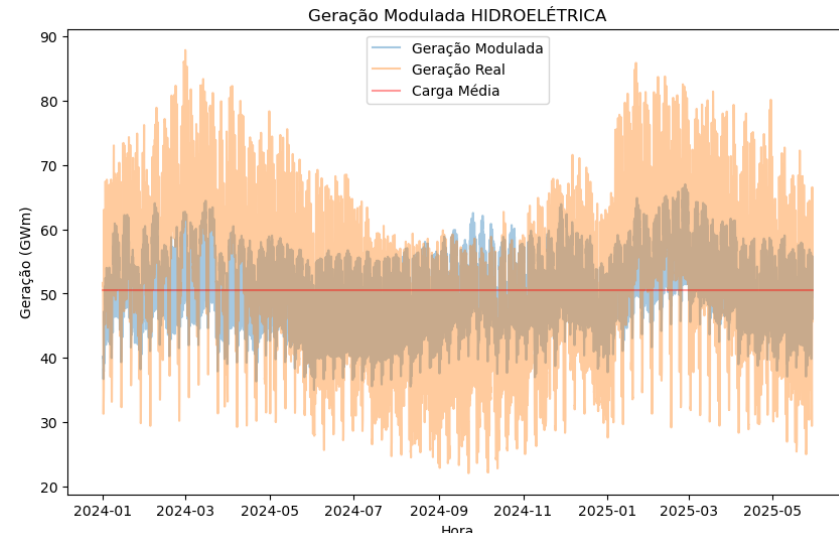
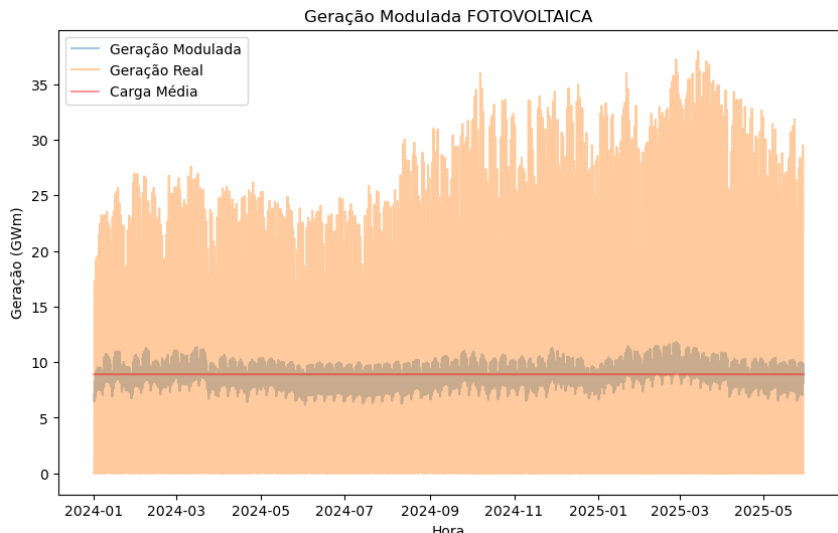
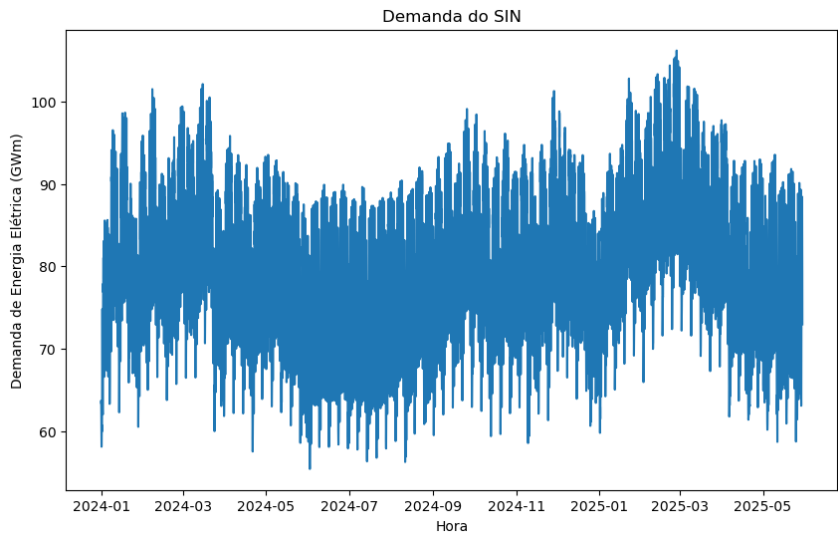
Spoiler!!

Fonte	Modulação (bi R\$)
FV	-2.700
Hidro	-5.240
Térmica	4.460
Eólica	3.230
Nuclear	-0.060

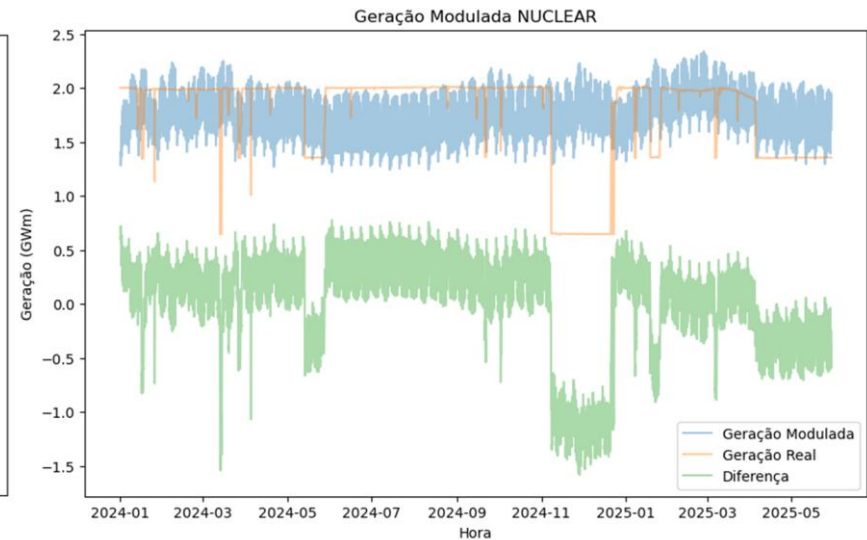
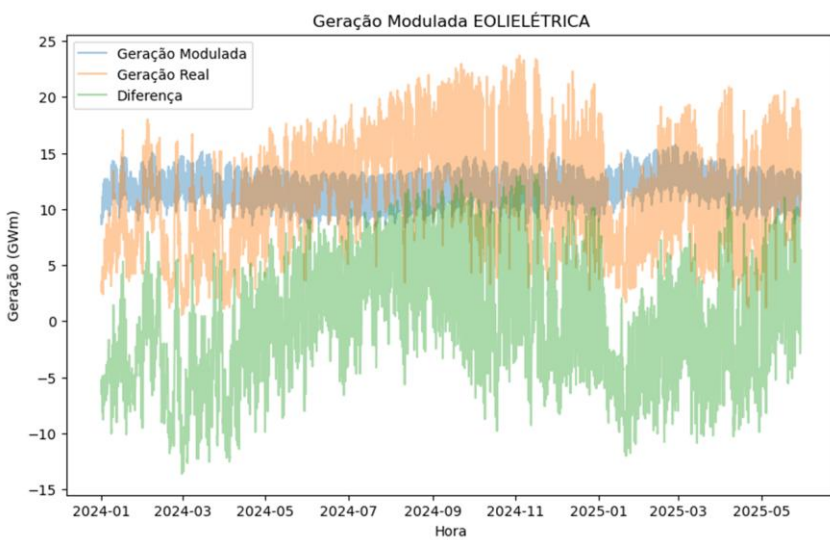
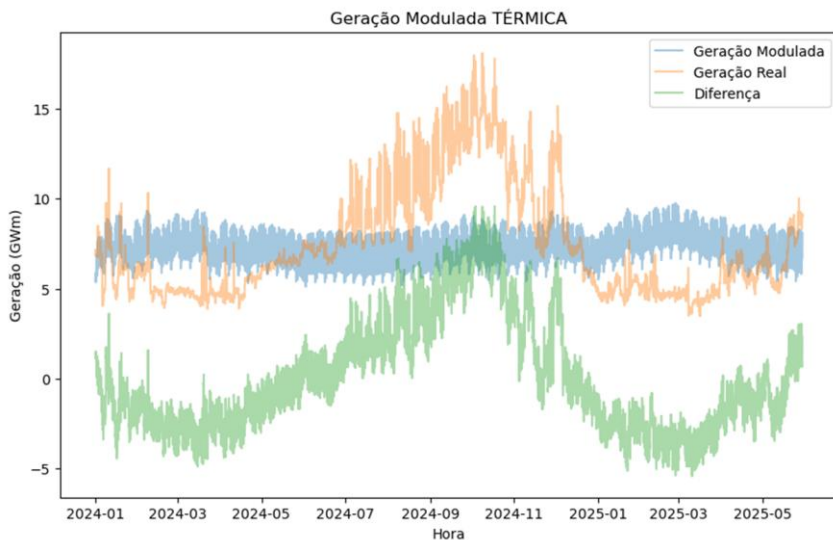
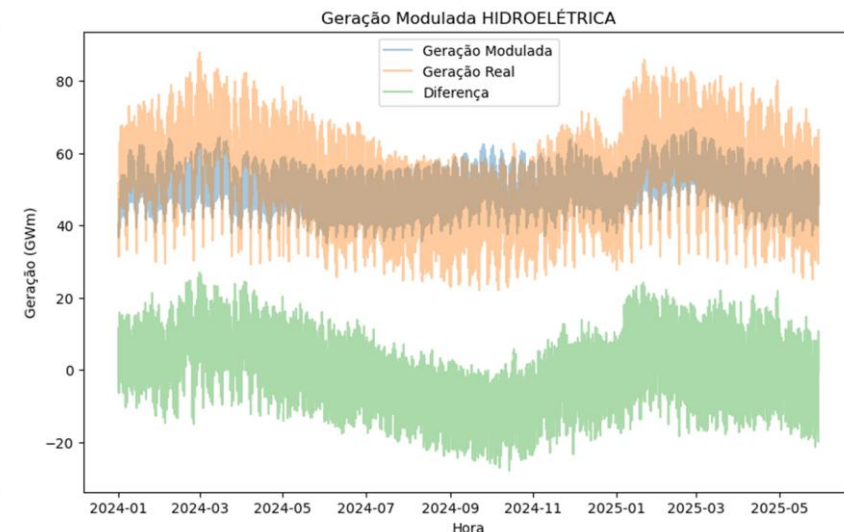
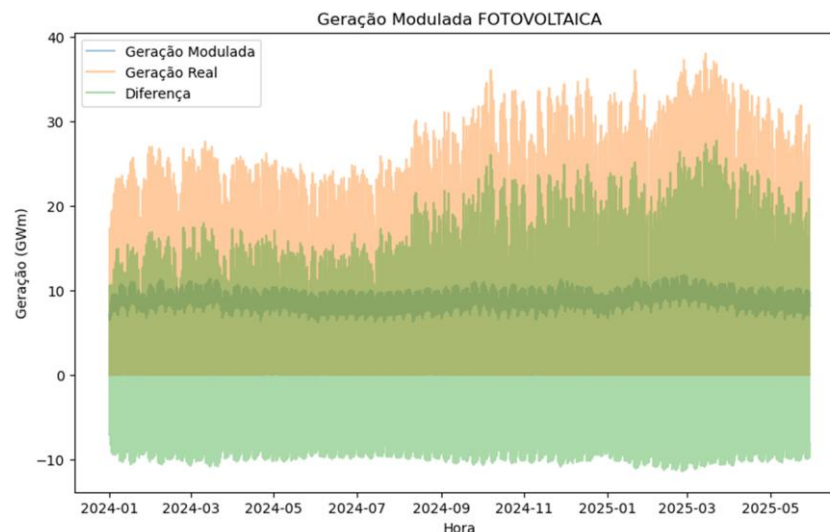
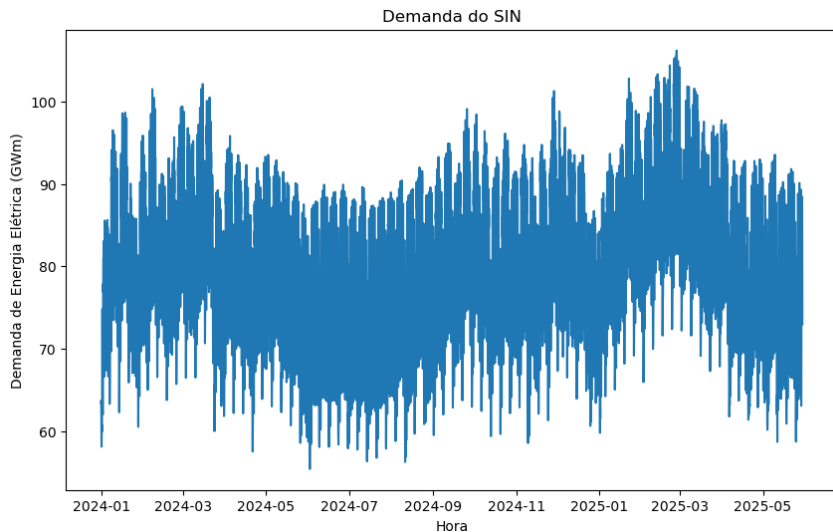
Metodologia



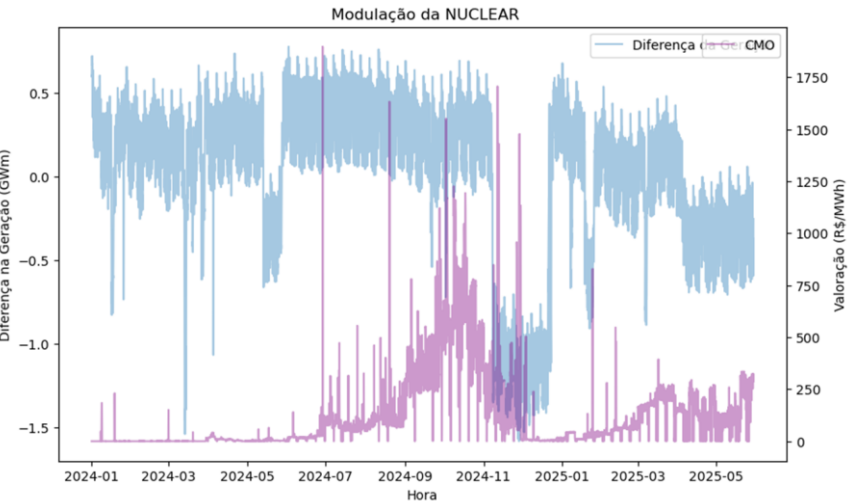
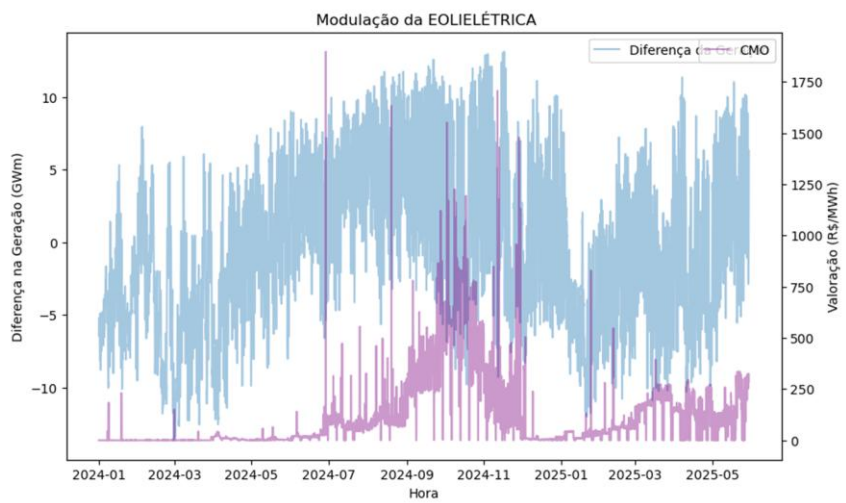
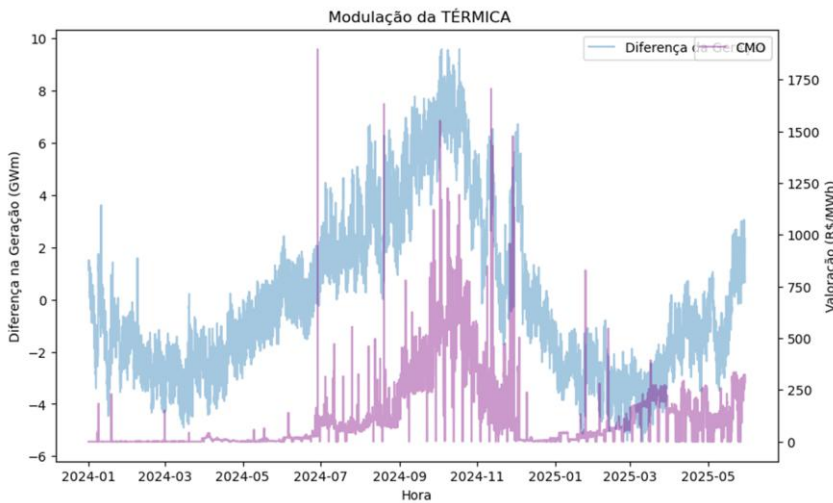
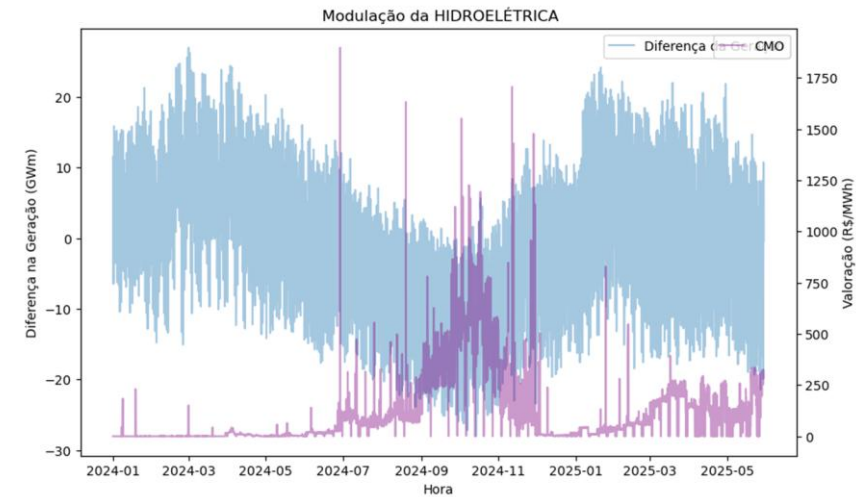
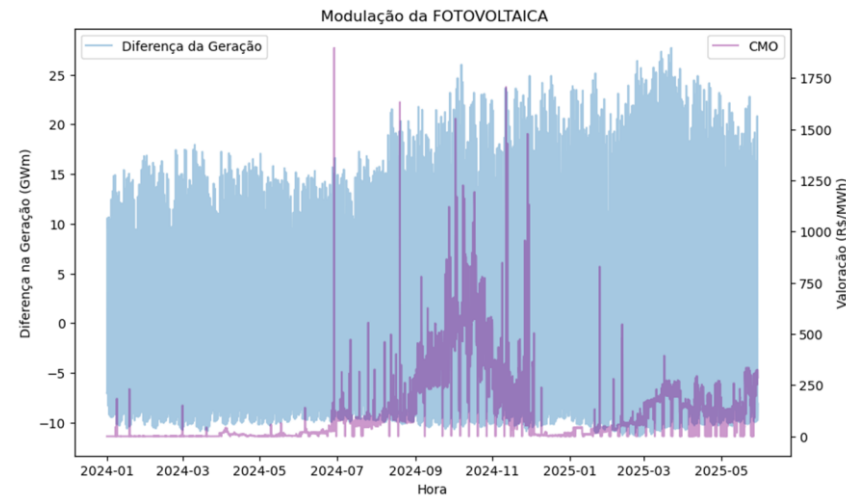
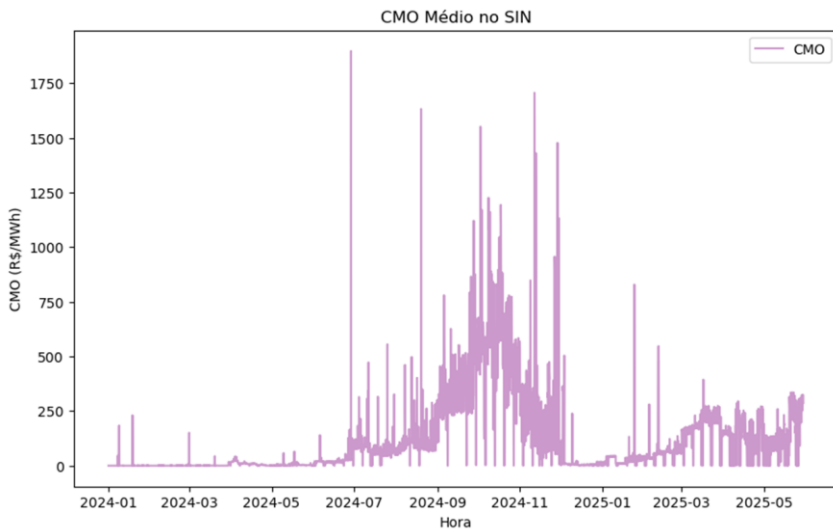
Geração Modulada



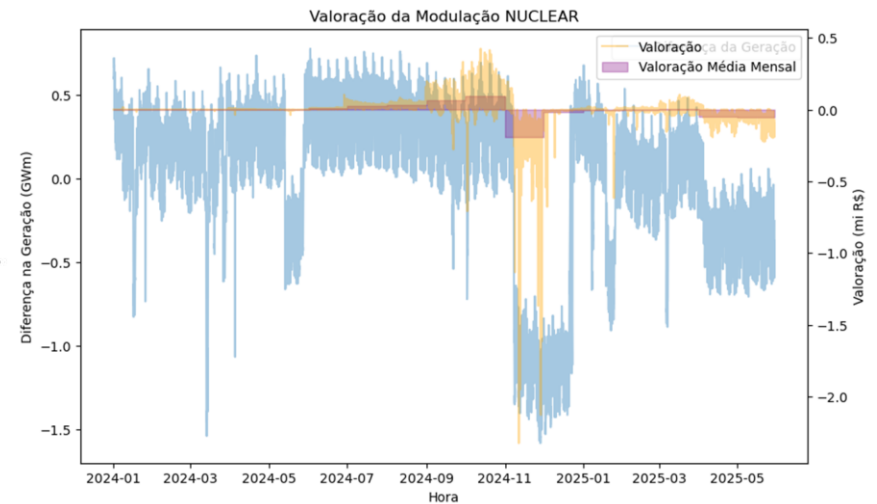
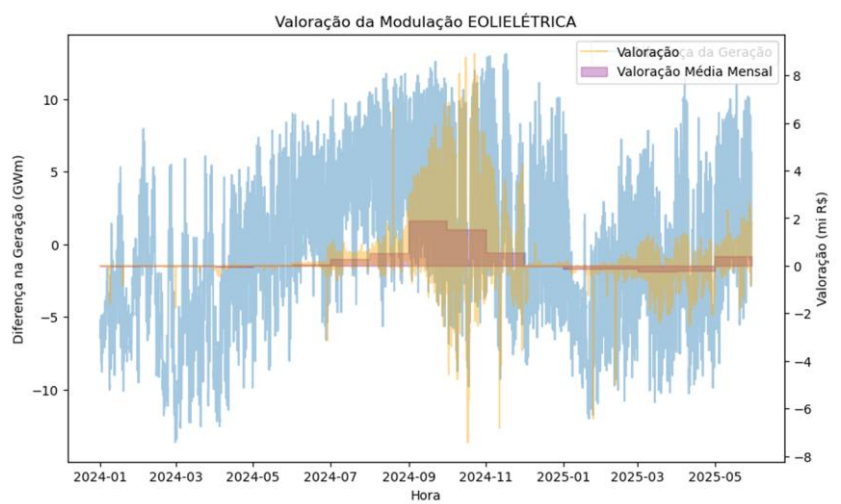
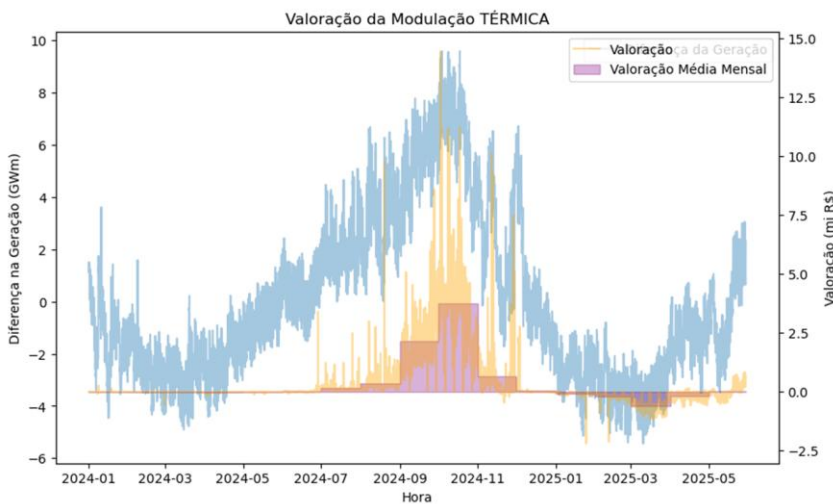
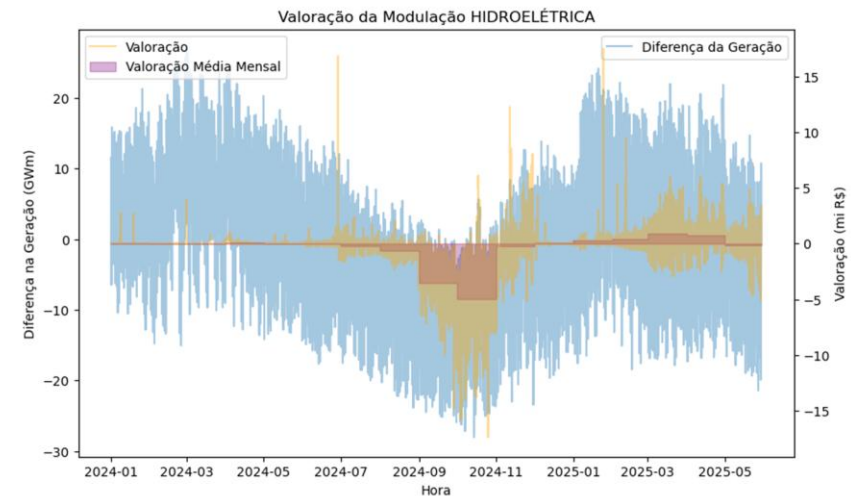
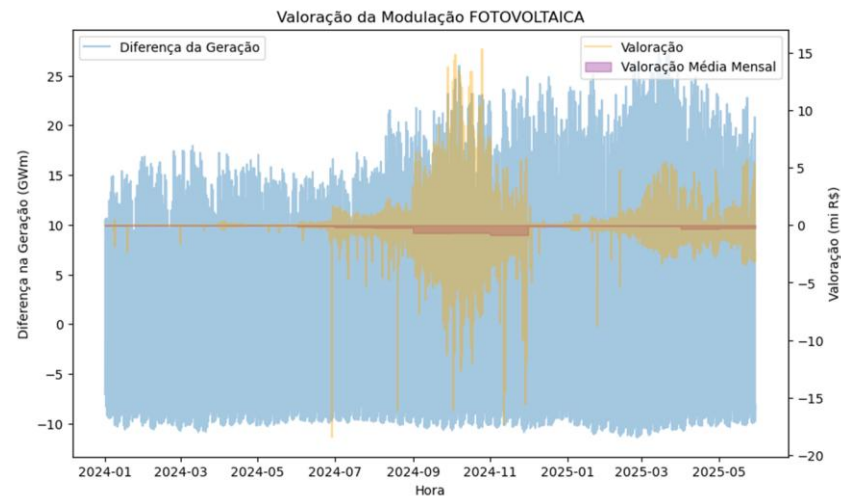
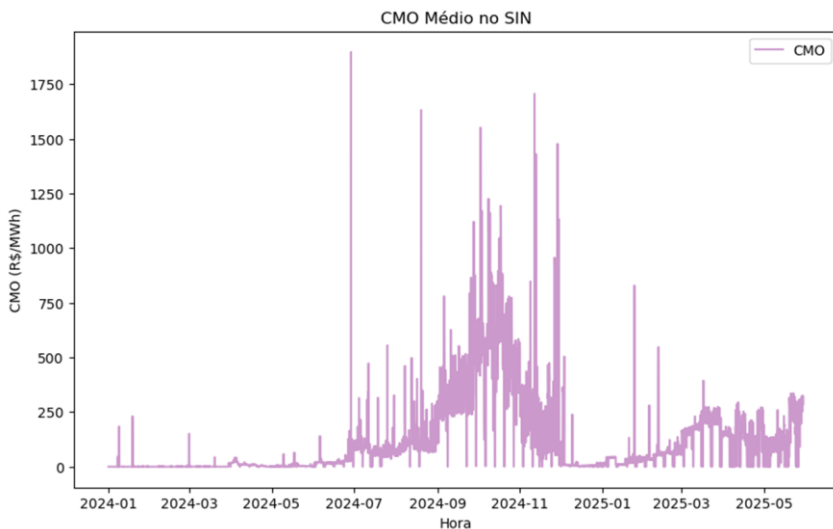
Diferença de Geração



CMO para valoração



Valoração da Modulação



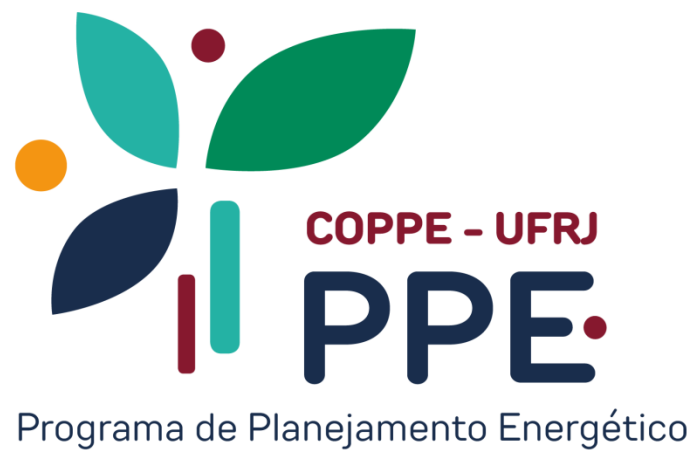
Conclusão

Resultados preliminares

Fonte	Modulação (bi R\$)
FV	-2.700
Hidro	-5.240
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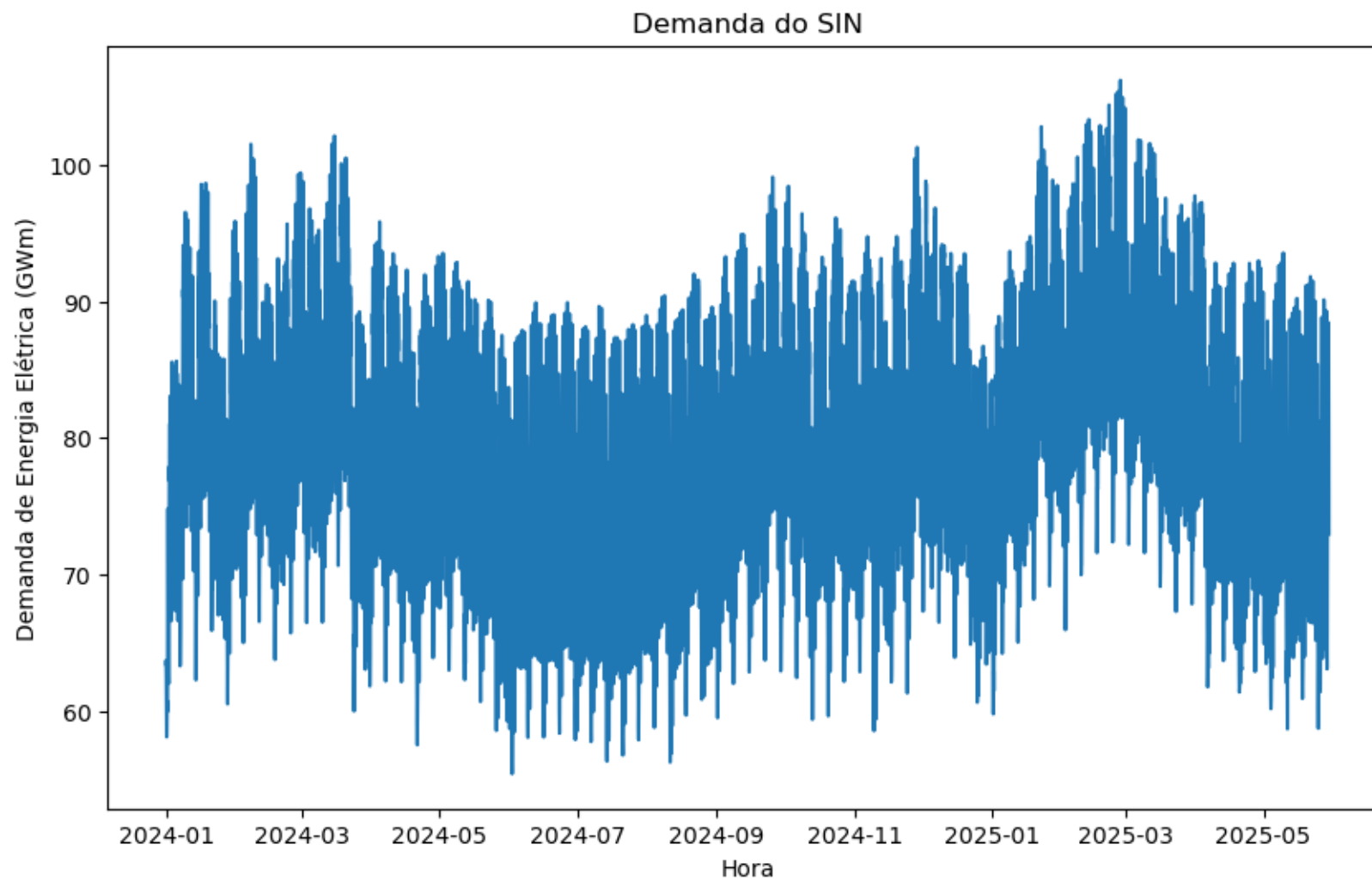
Atividades Futuras

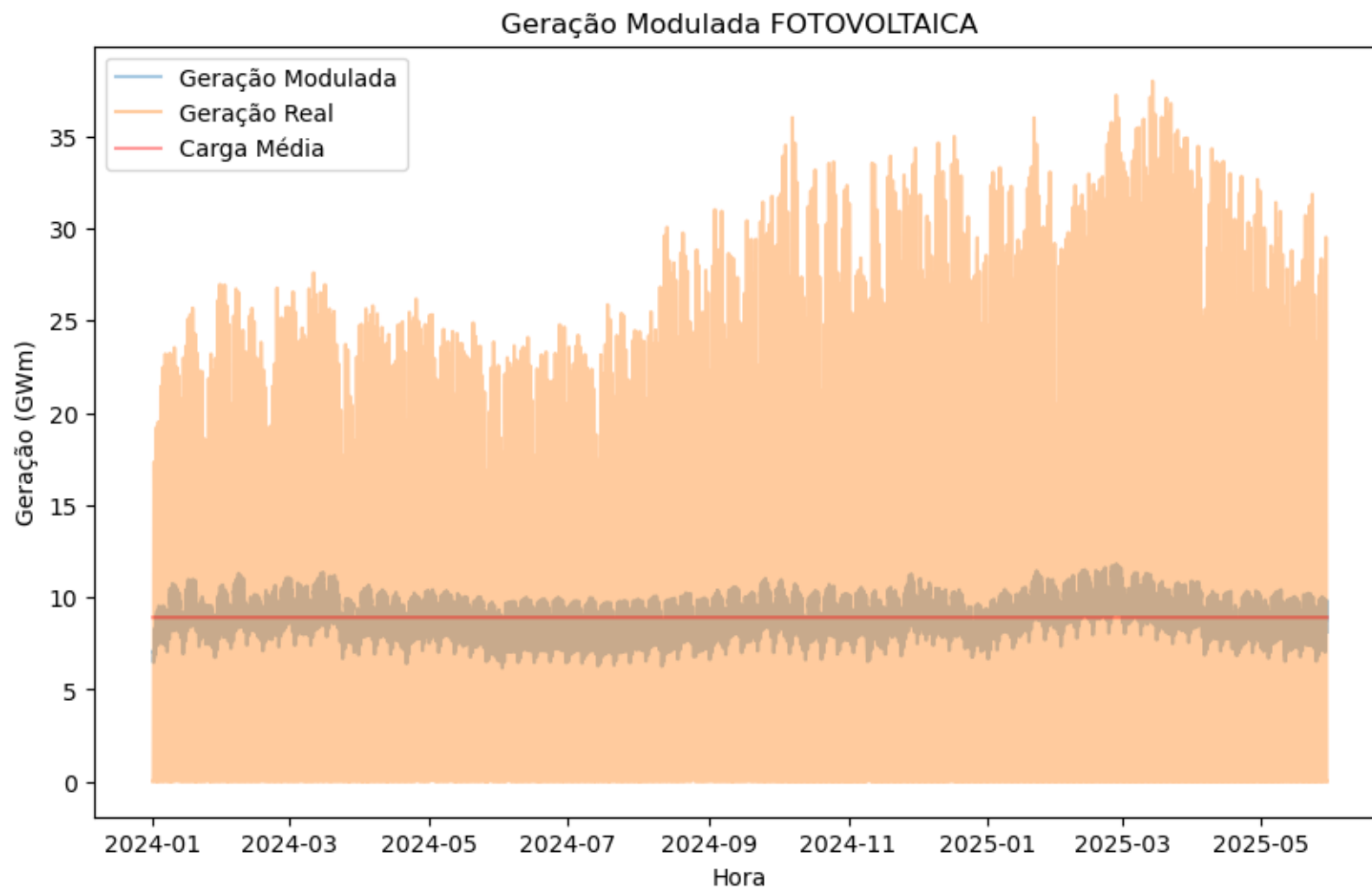
- Fazer modulação em períodos menores (6 em 6 meses) e avaliar a modulação semanal (eólica e UHE)
- Avaliar modulação em quarto de dia (solar)
- Considerar os intercâmbios
- Modular por subsistema
- Adquirir dados de geração máxima e geração mínima para avaliar violações

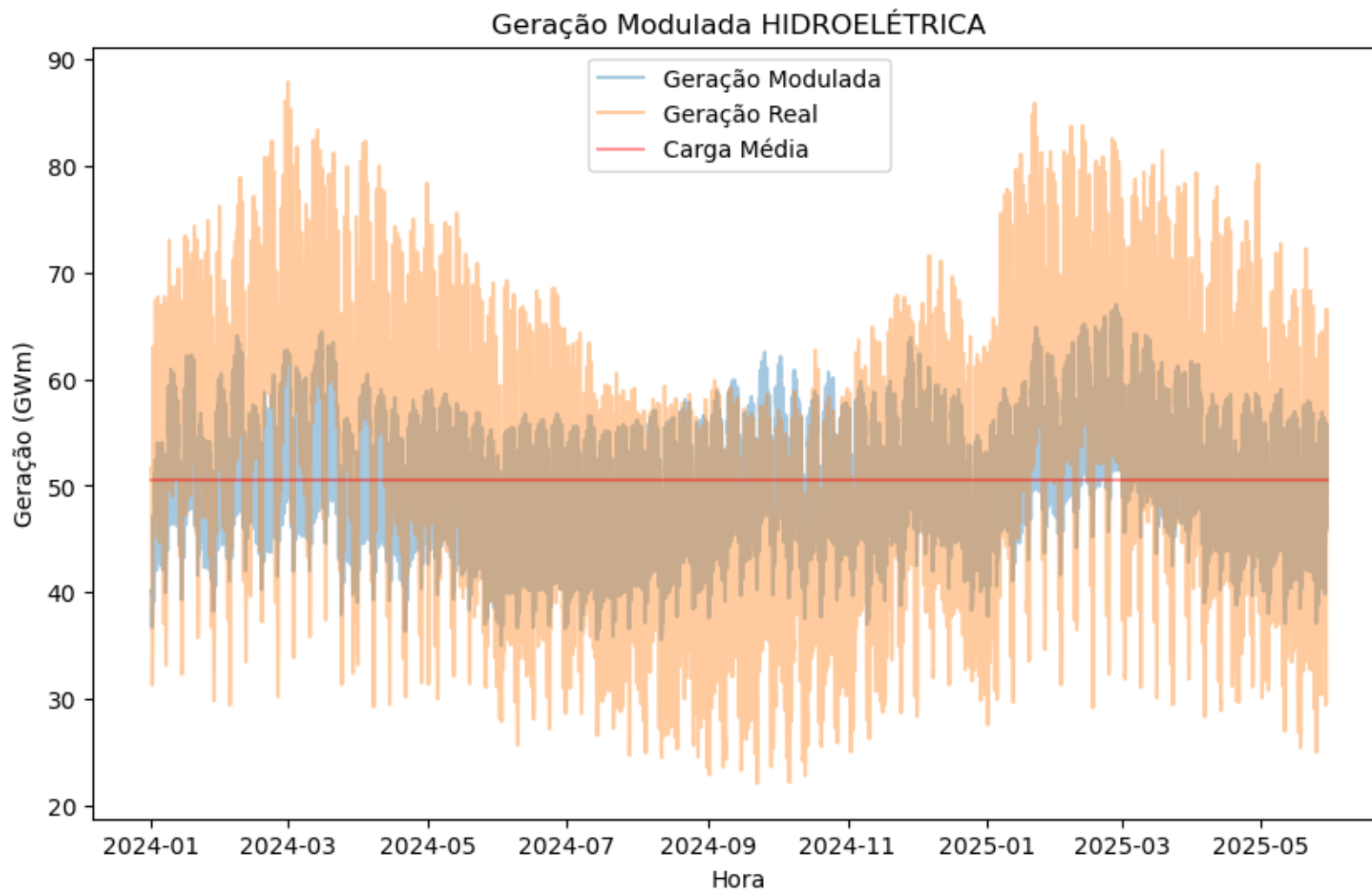


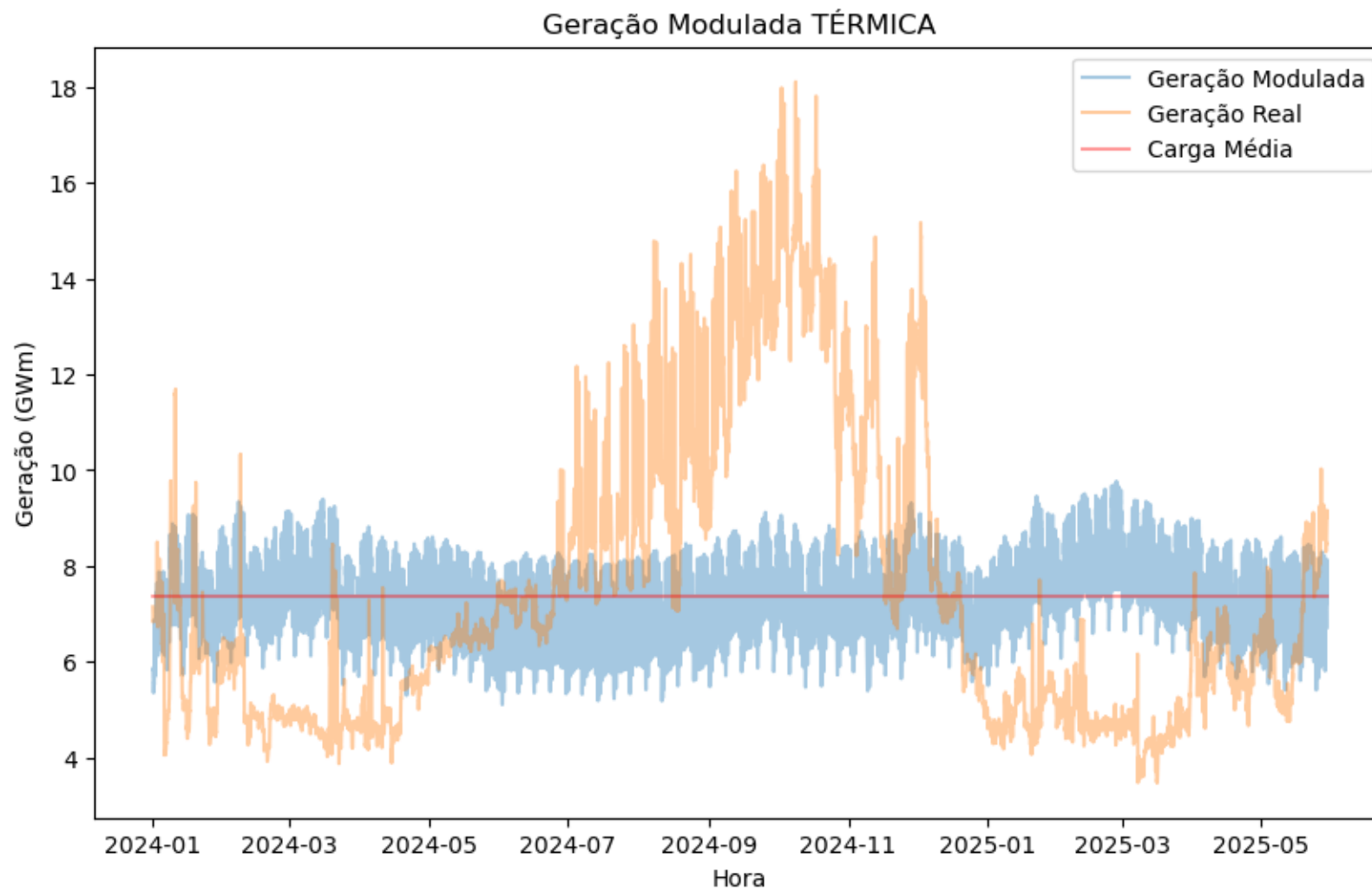
Av. Horácio de Macedo, 2030, Centro de Tecnologia - C 211
Cidade Universitária . Ilha do Fundão, RJ - CEP 21.941-914

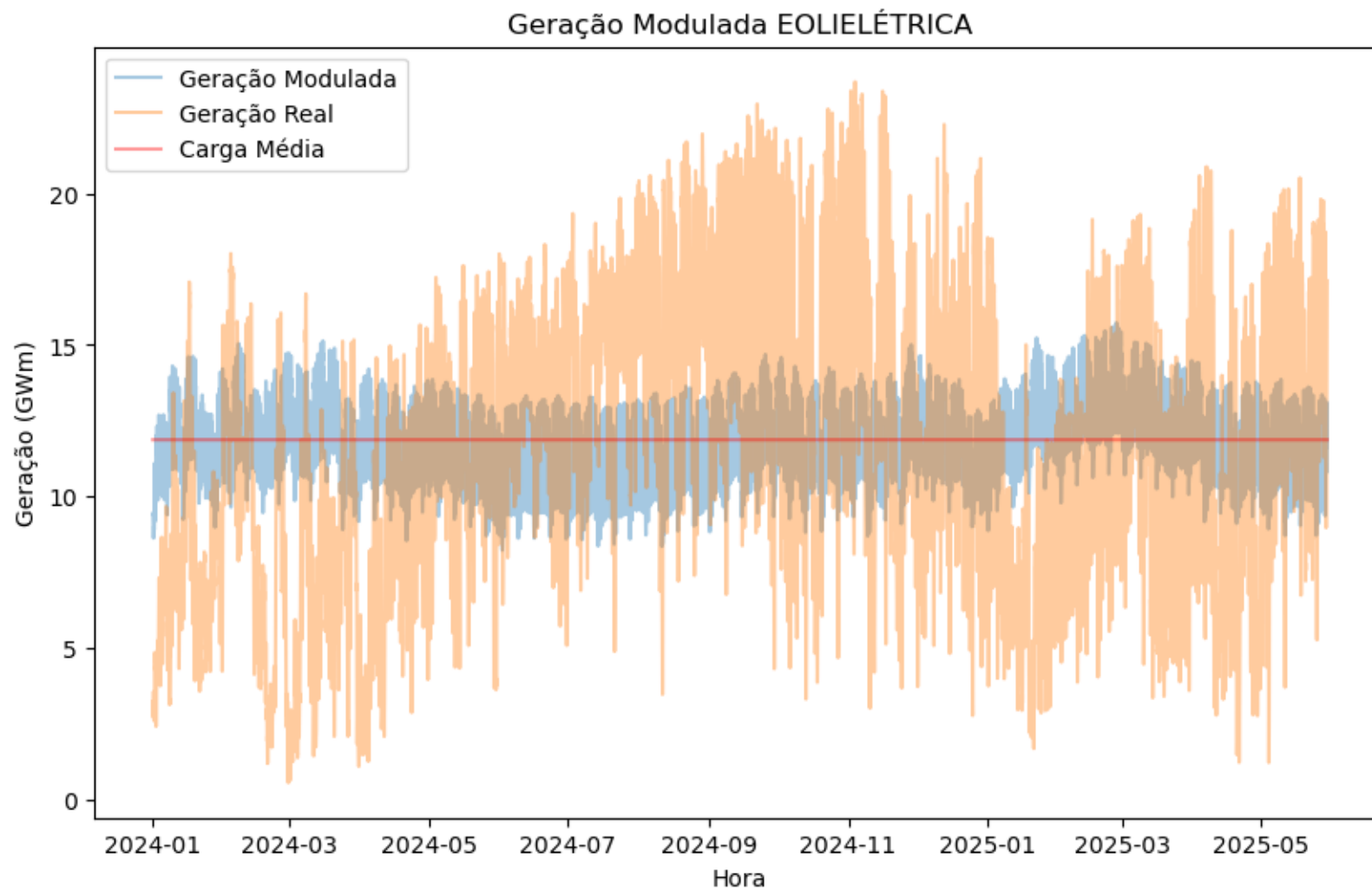
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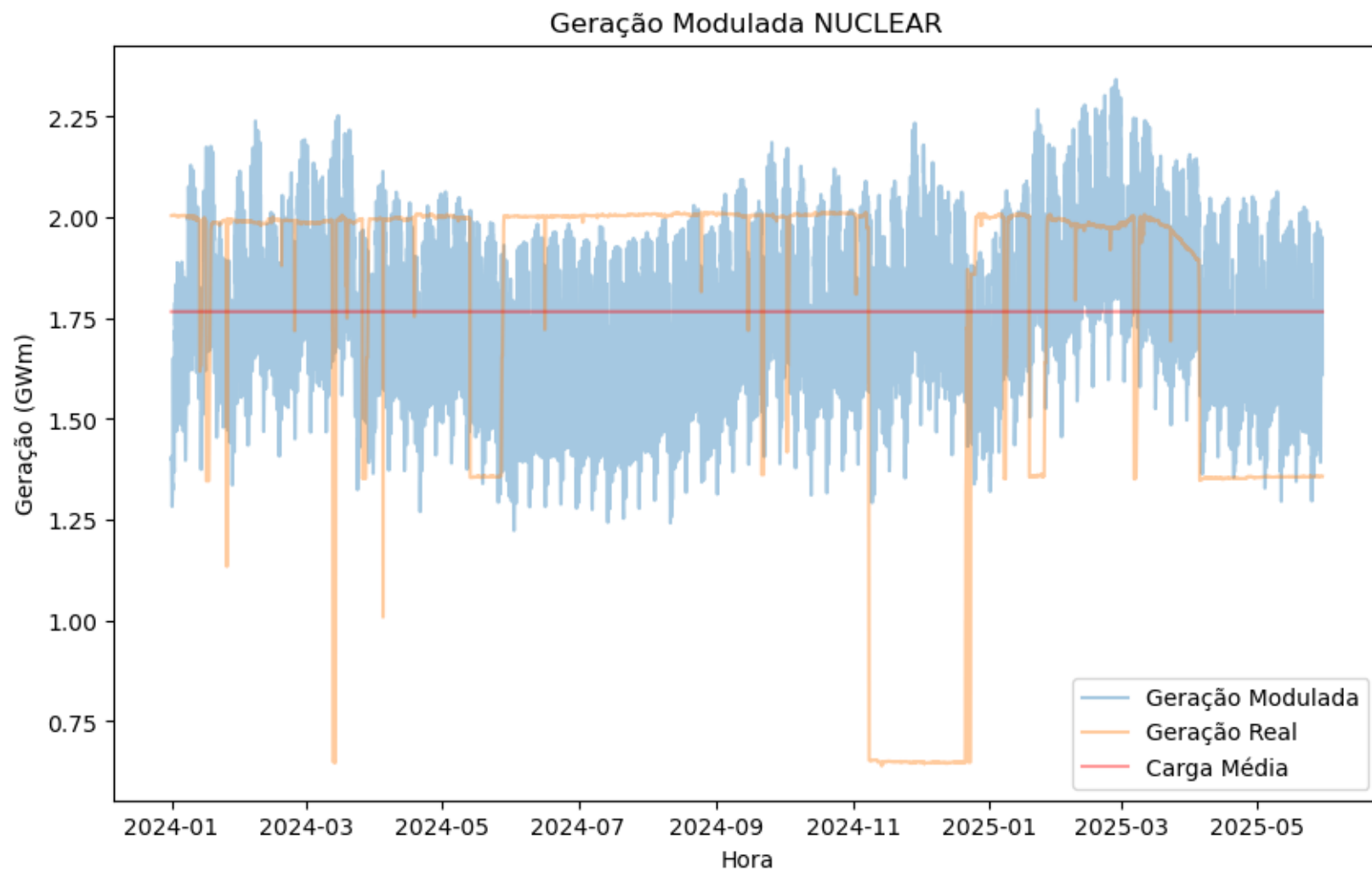


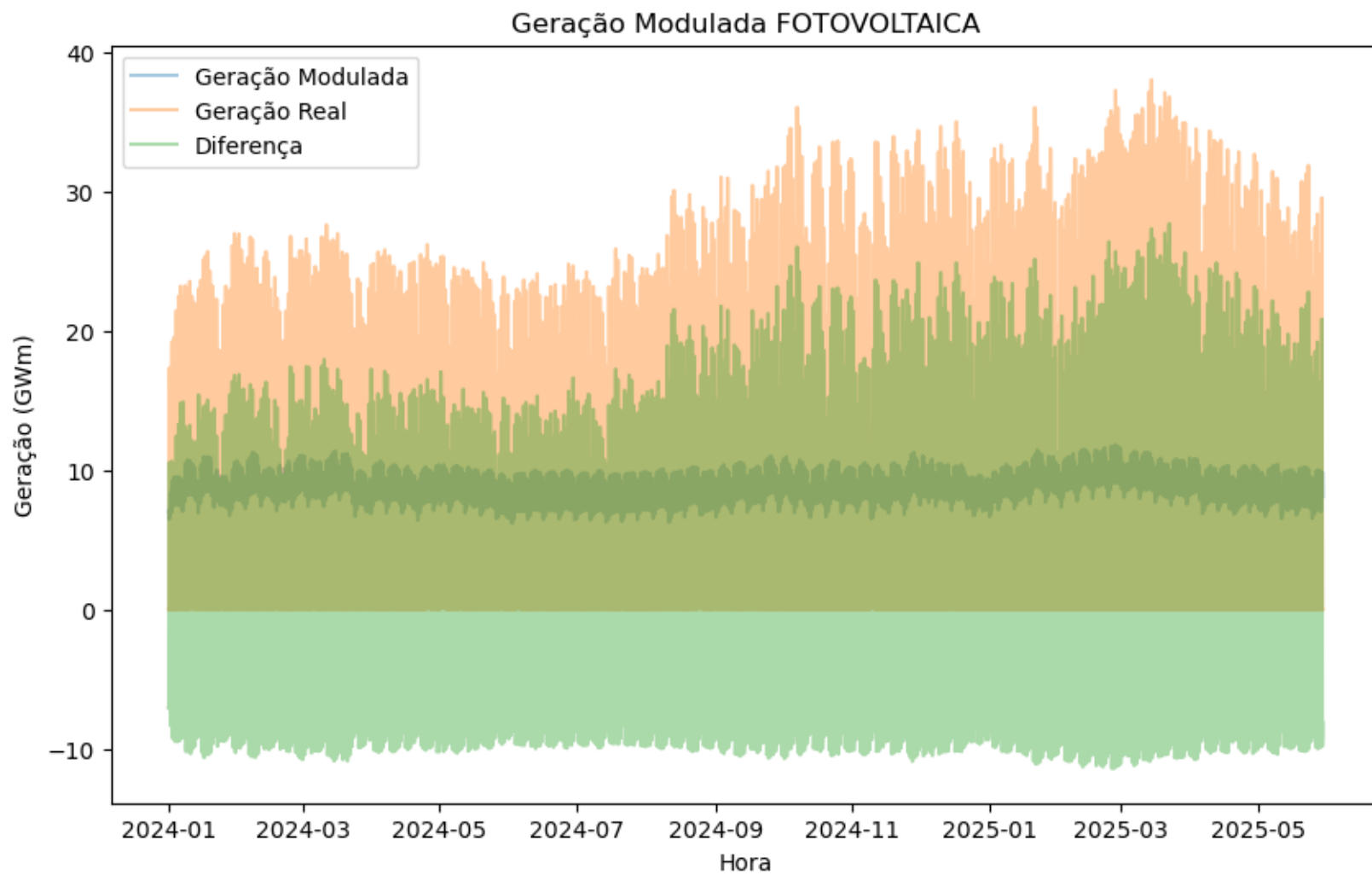


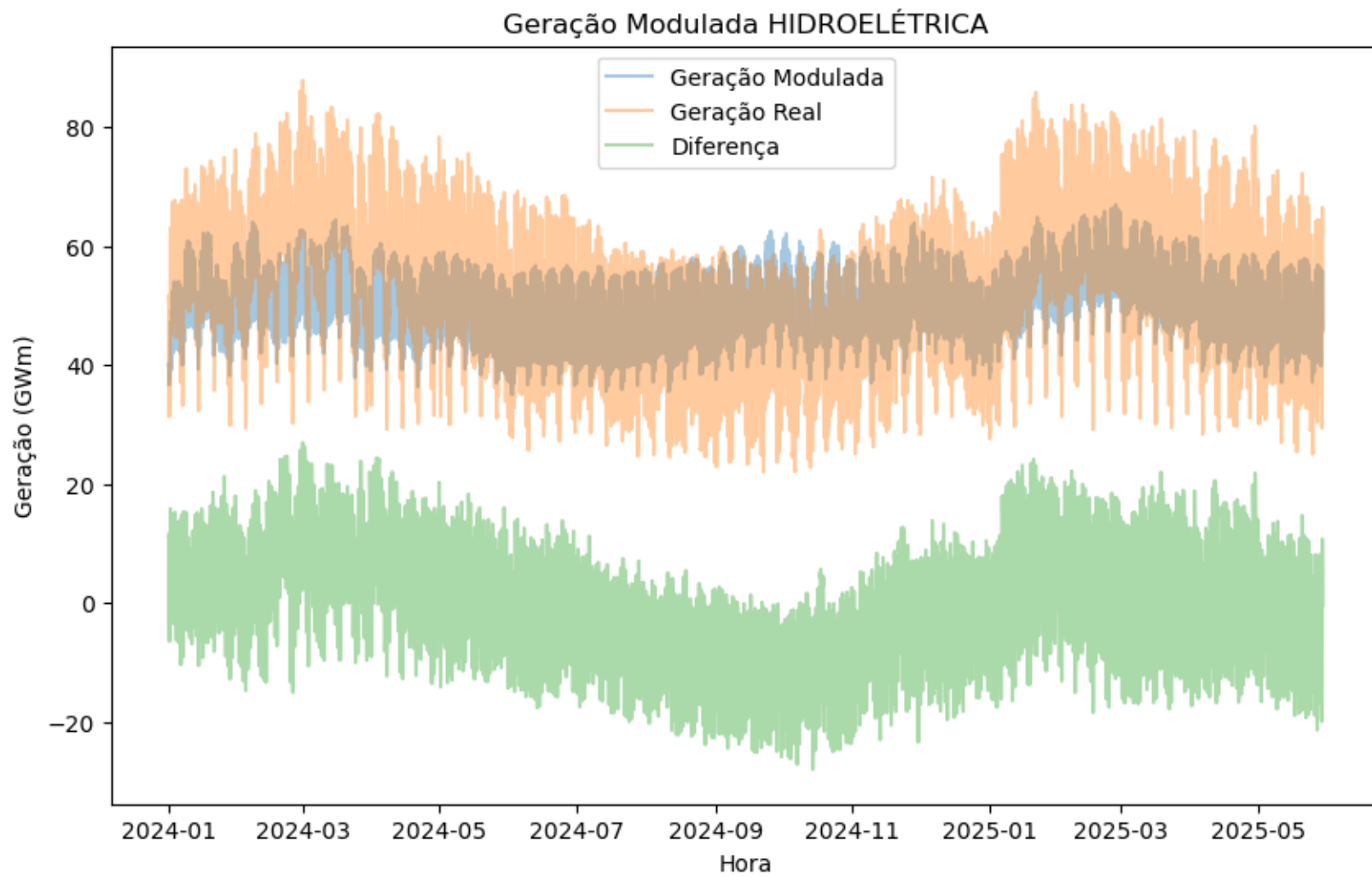


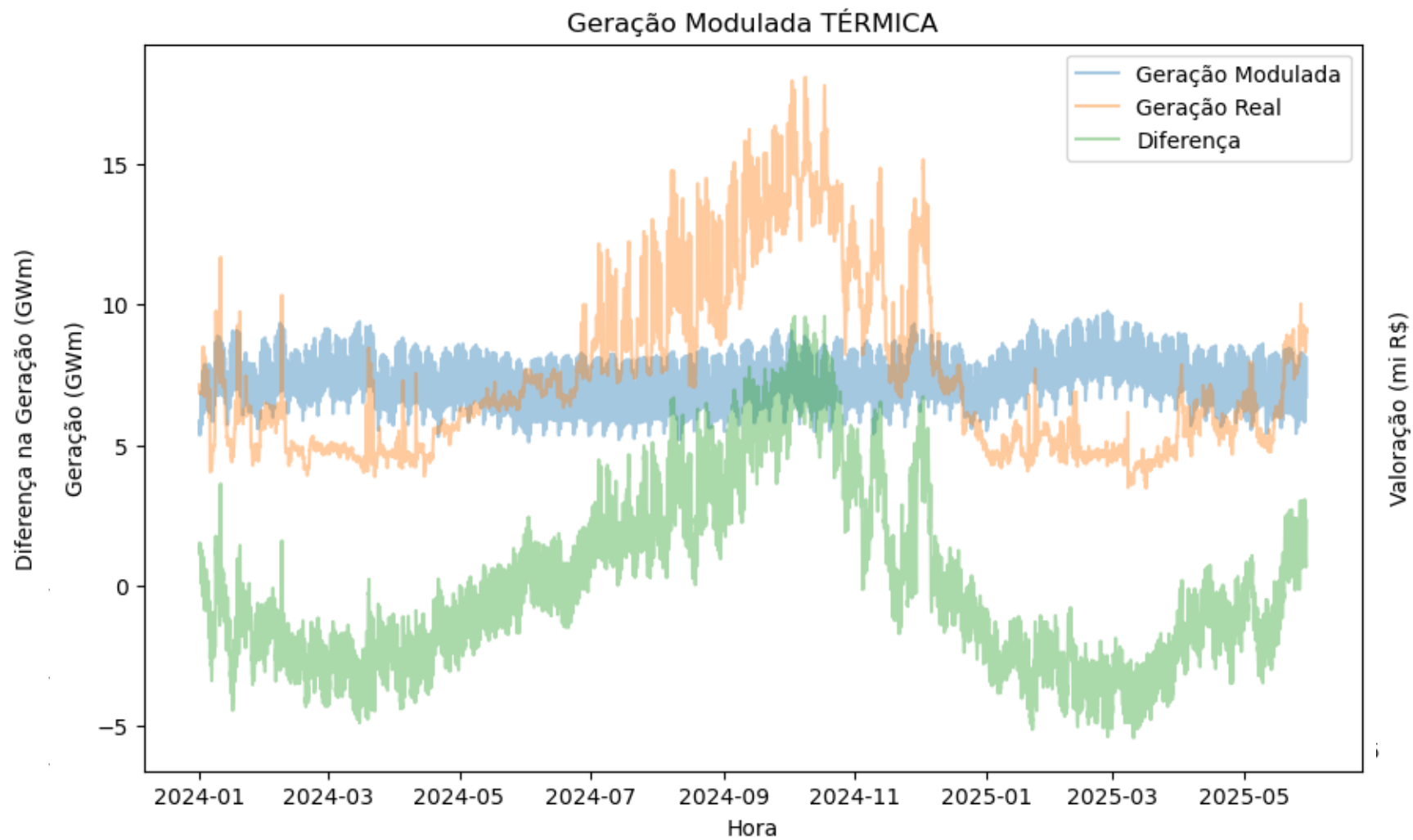


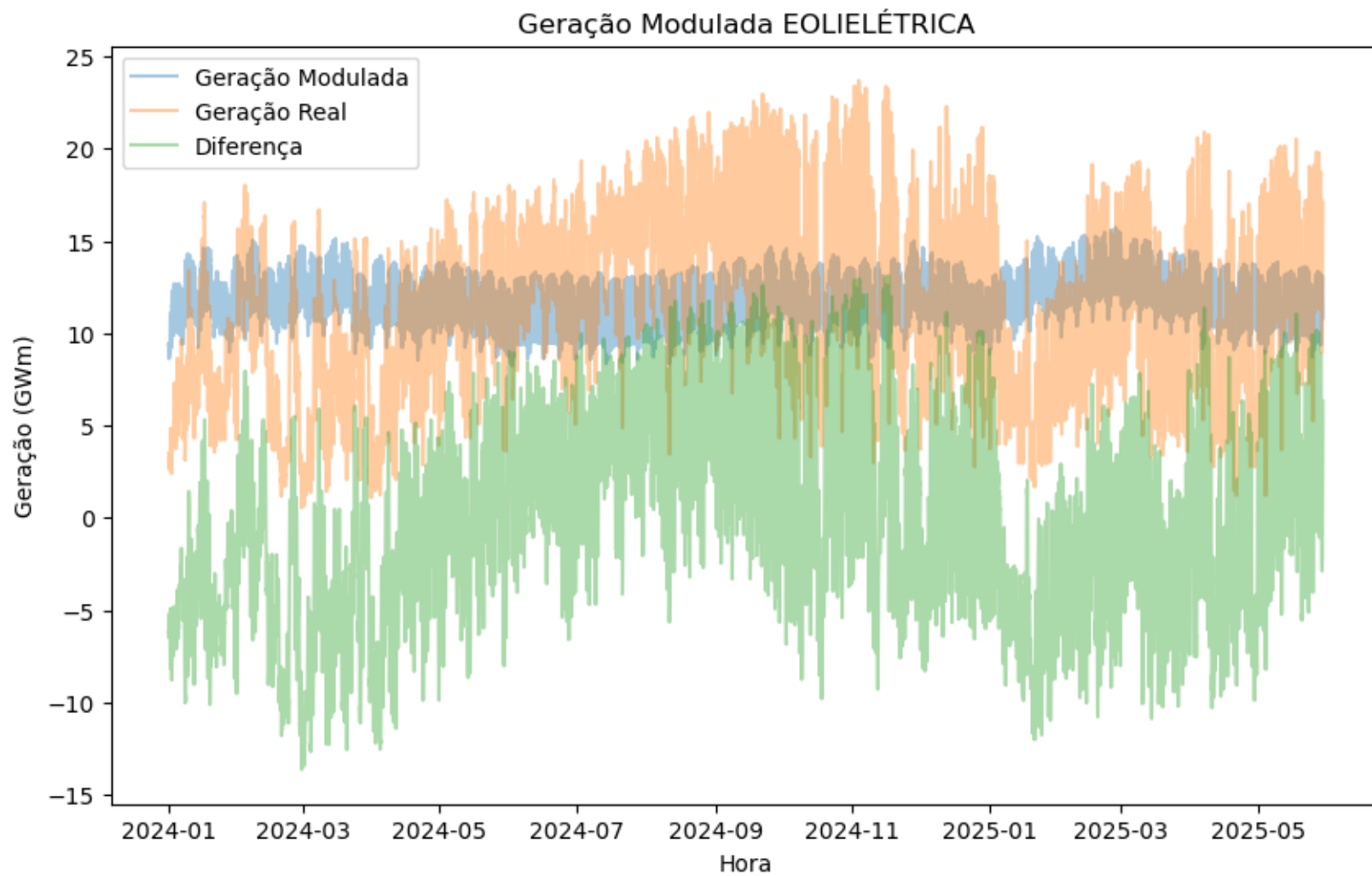


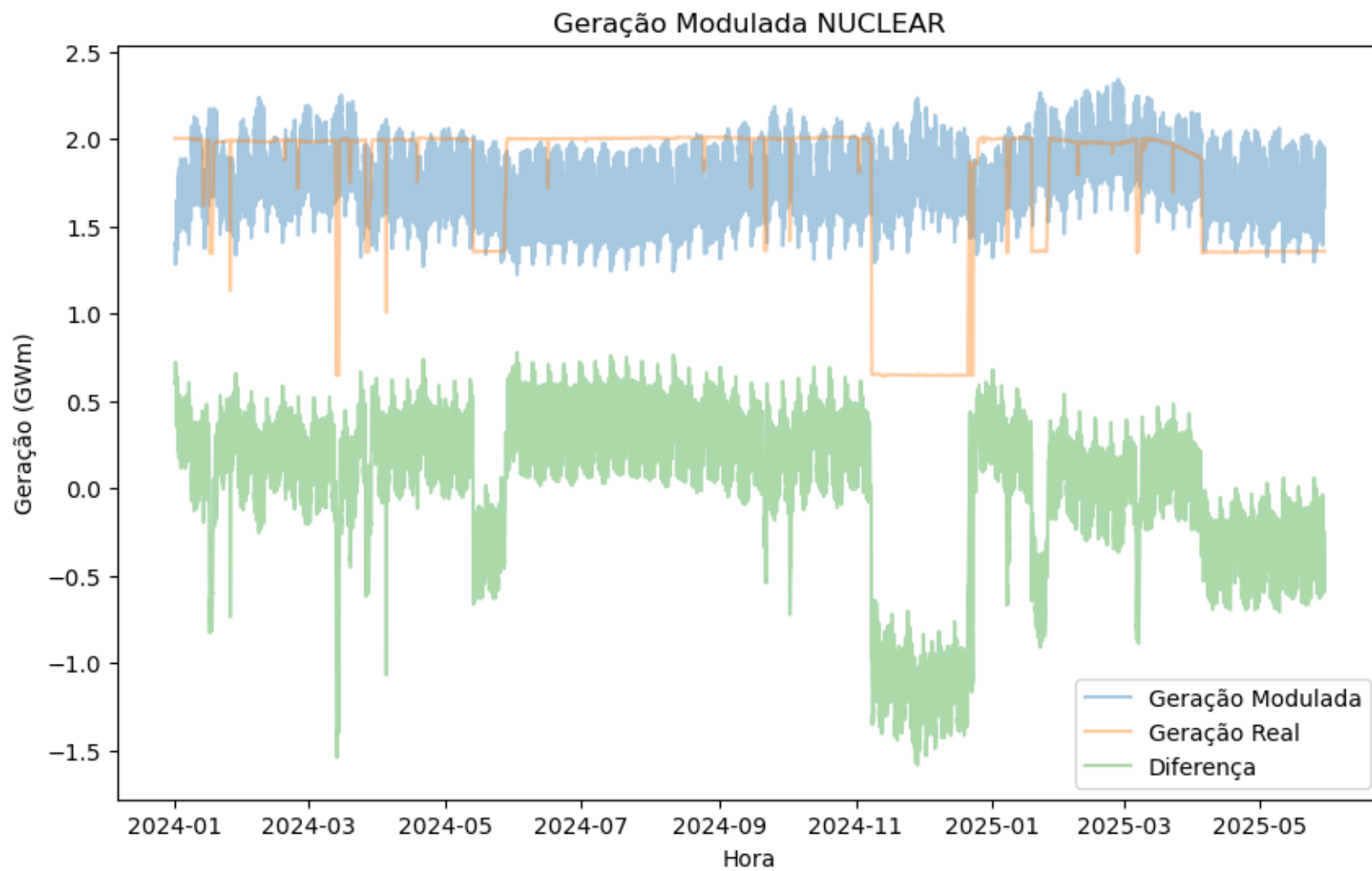


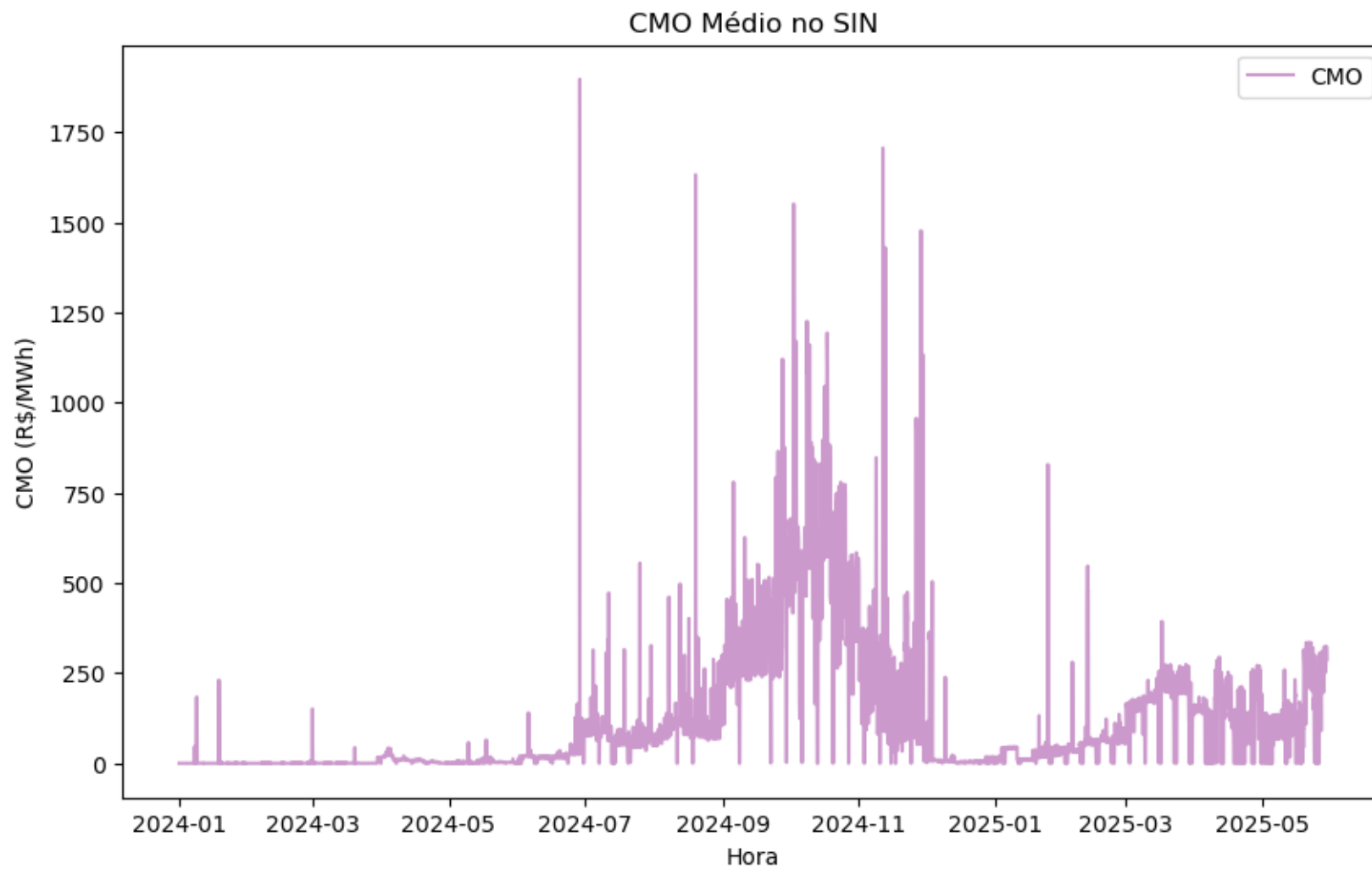


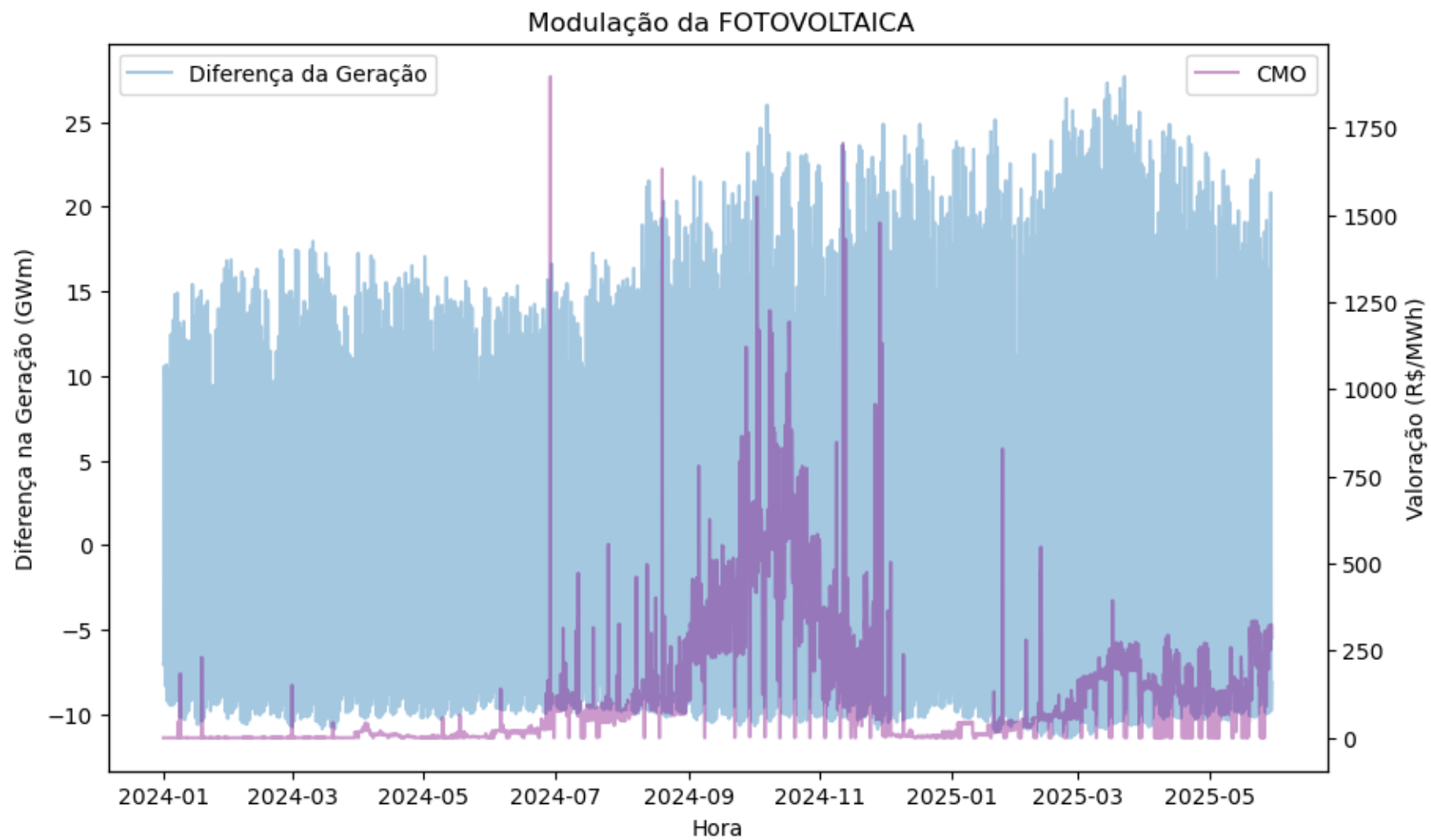


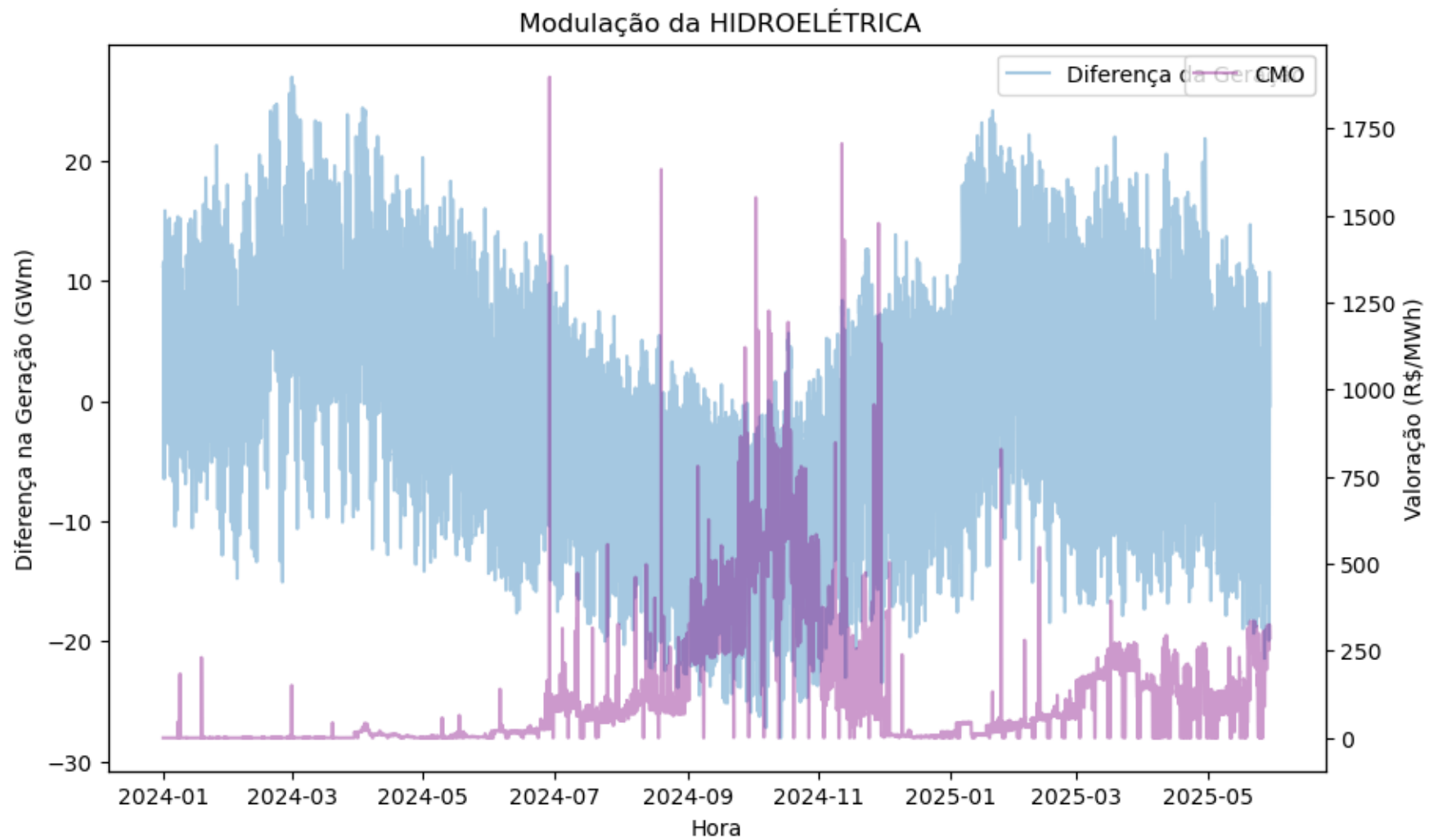


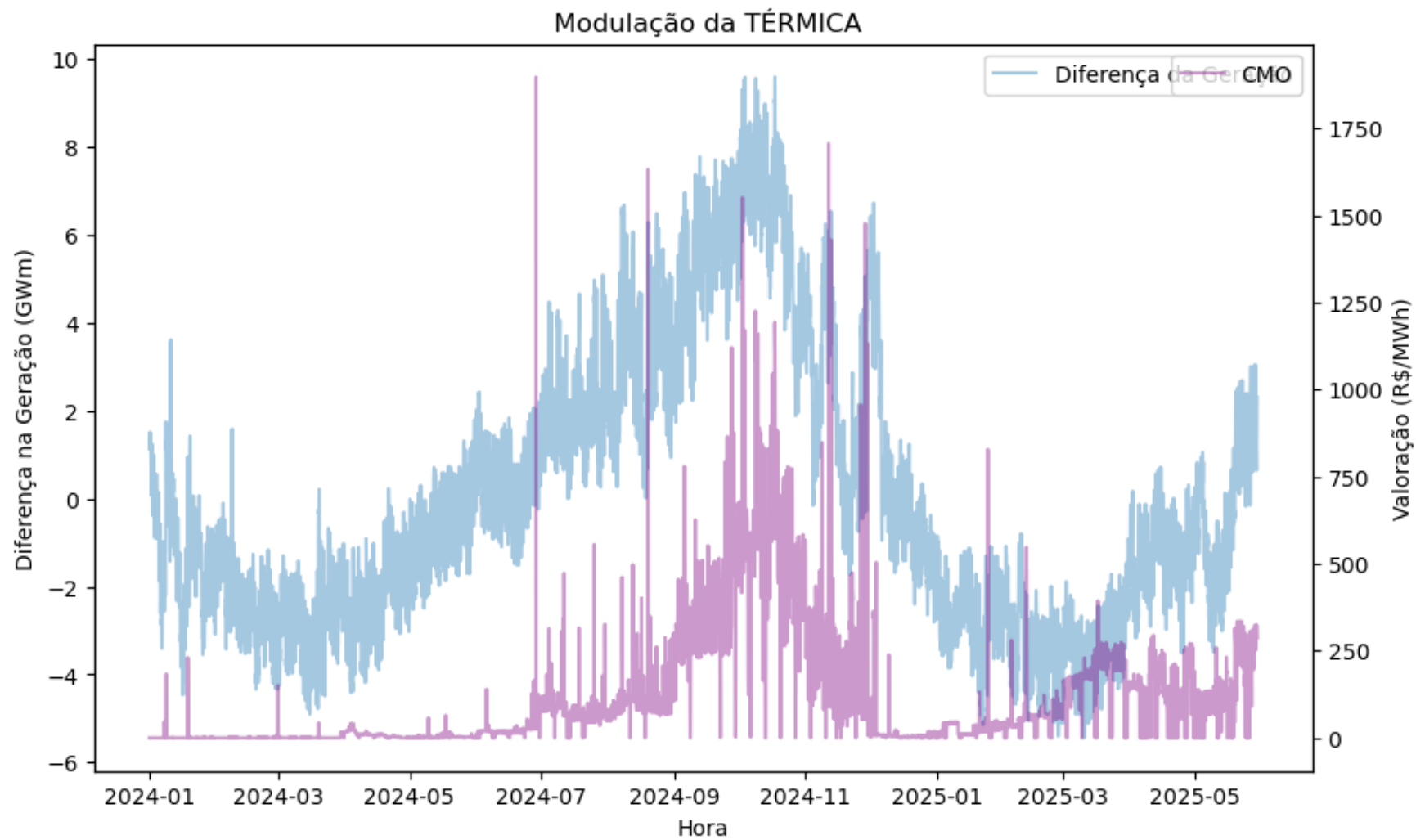


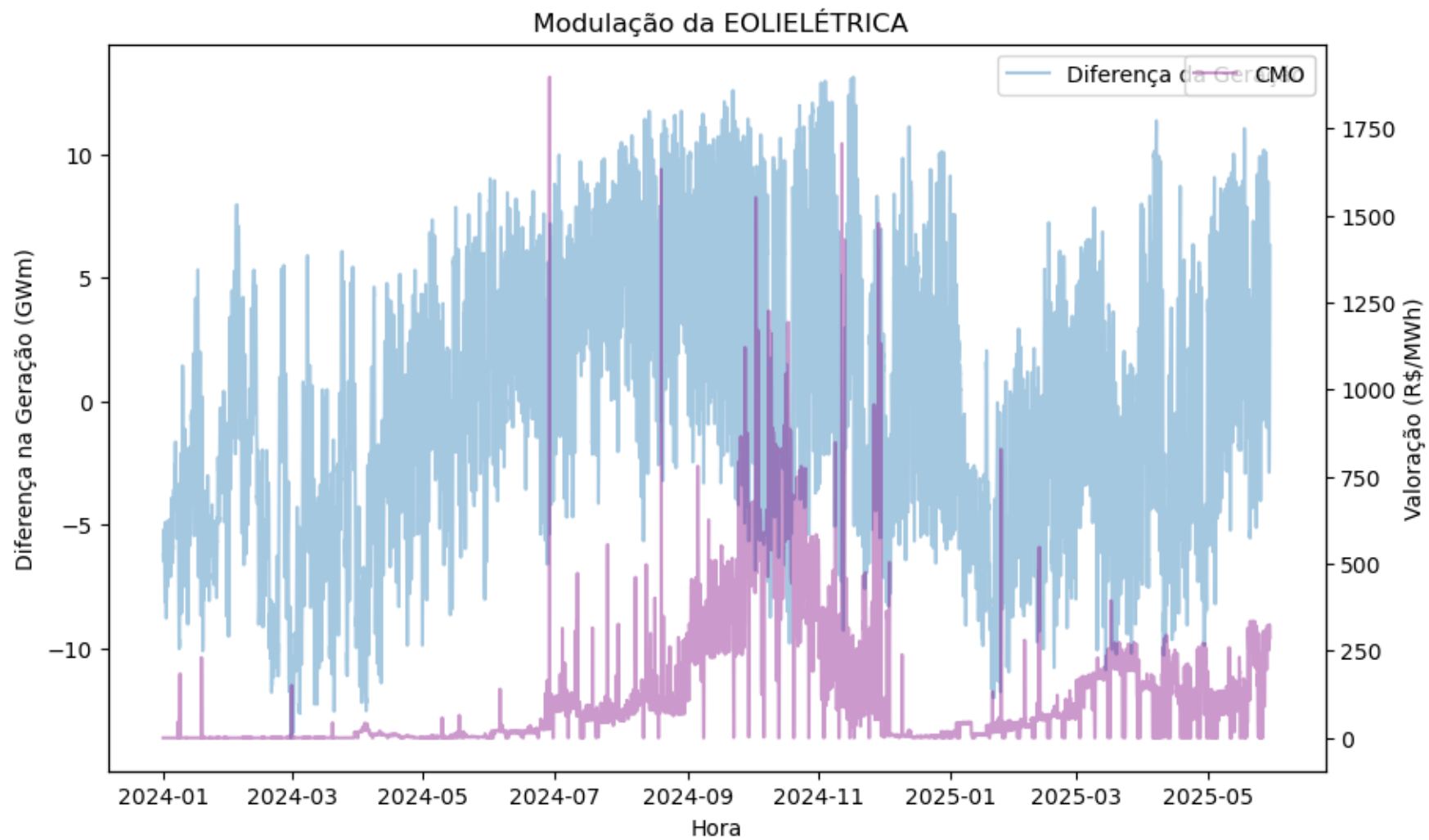


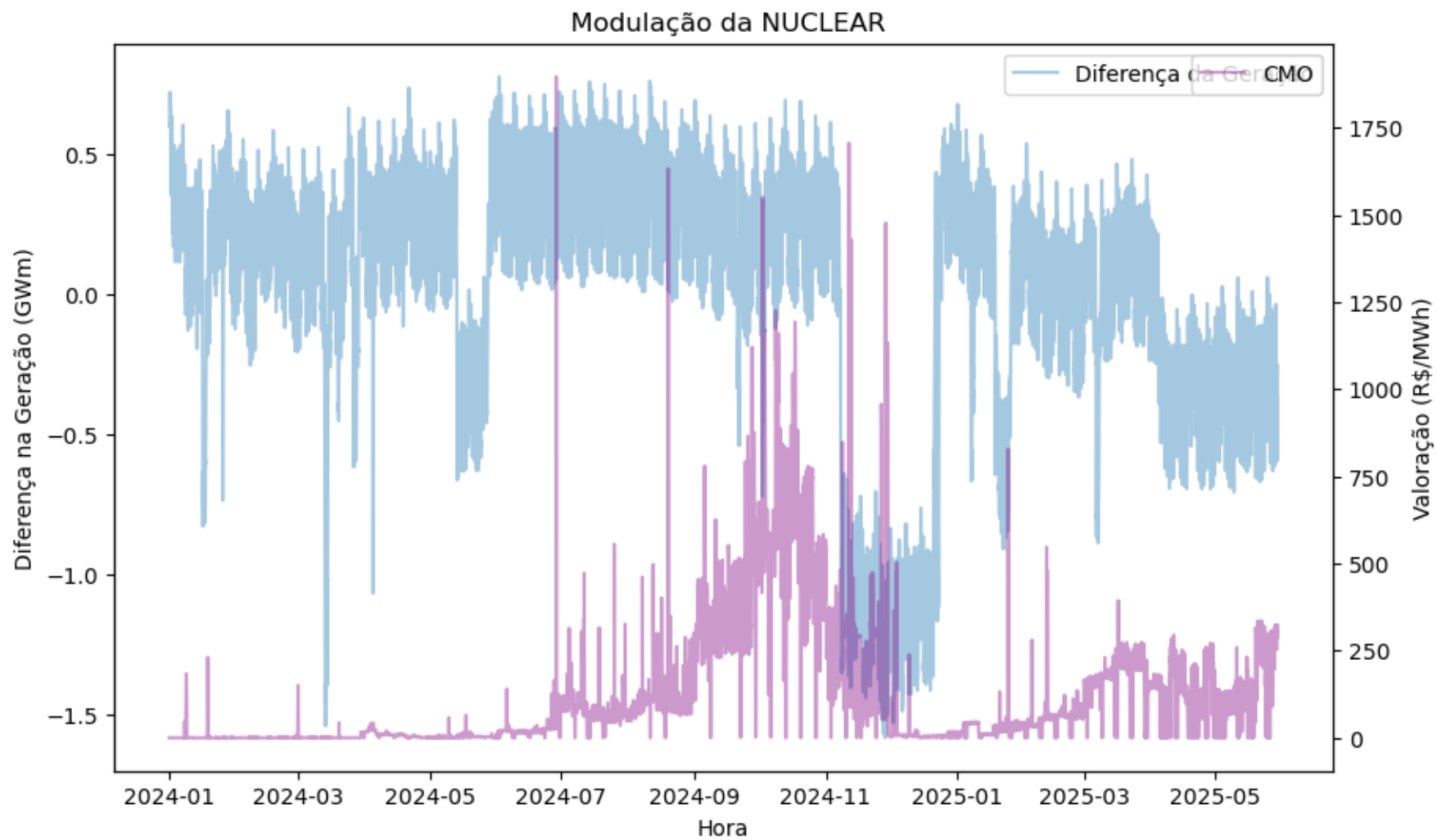


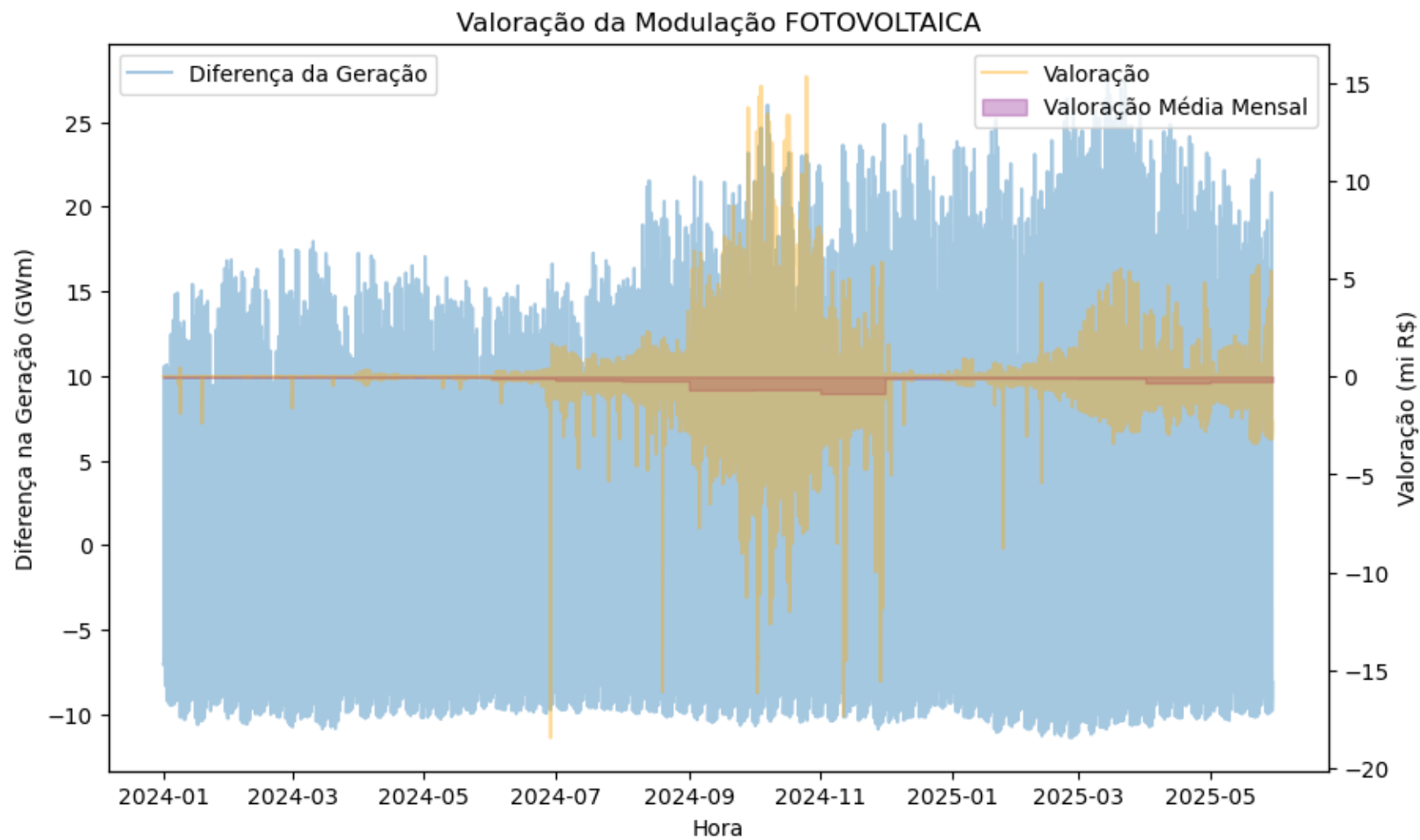


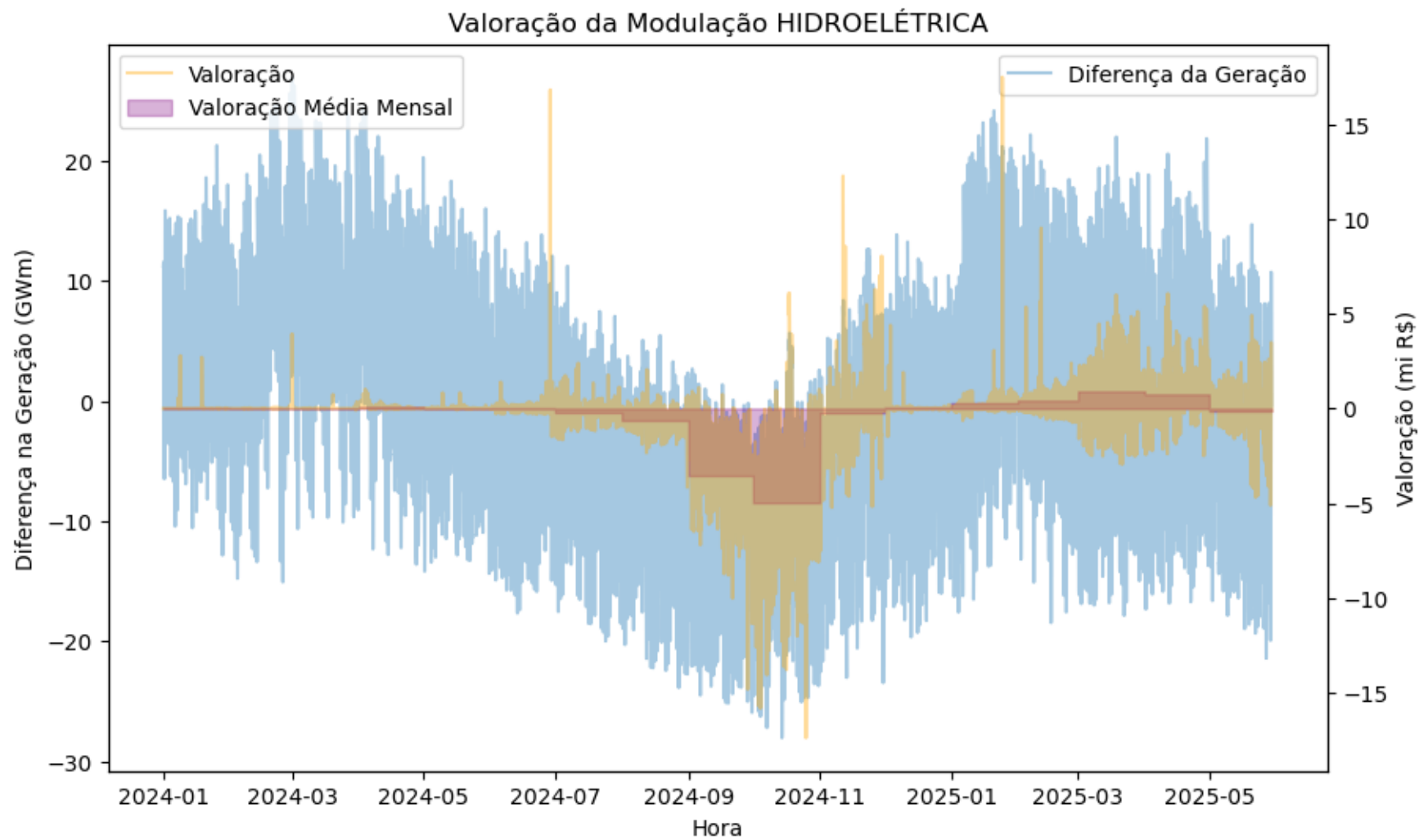


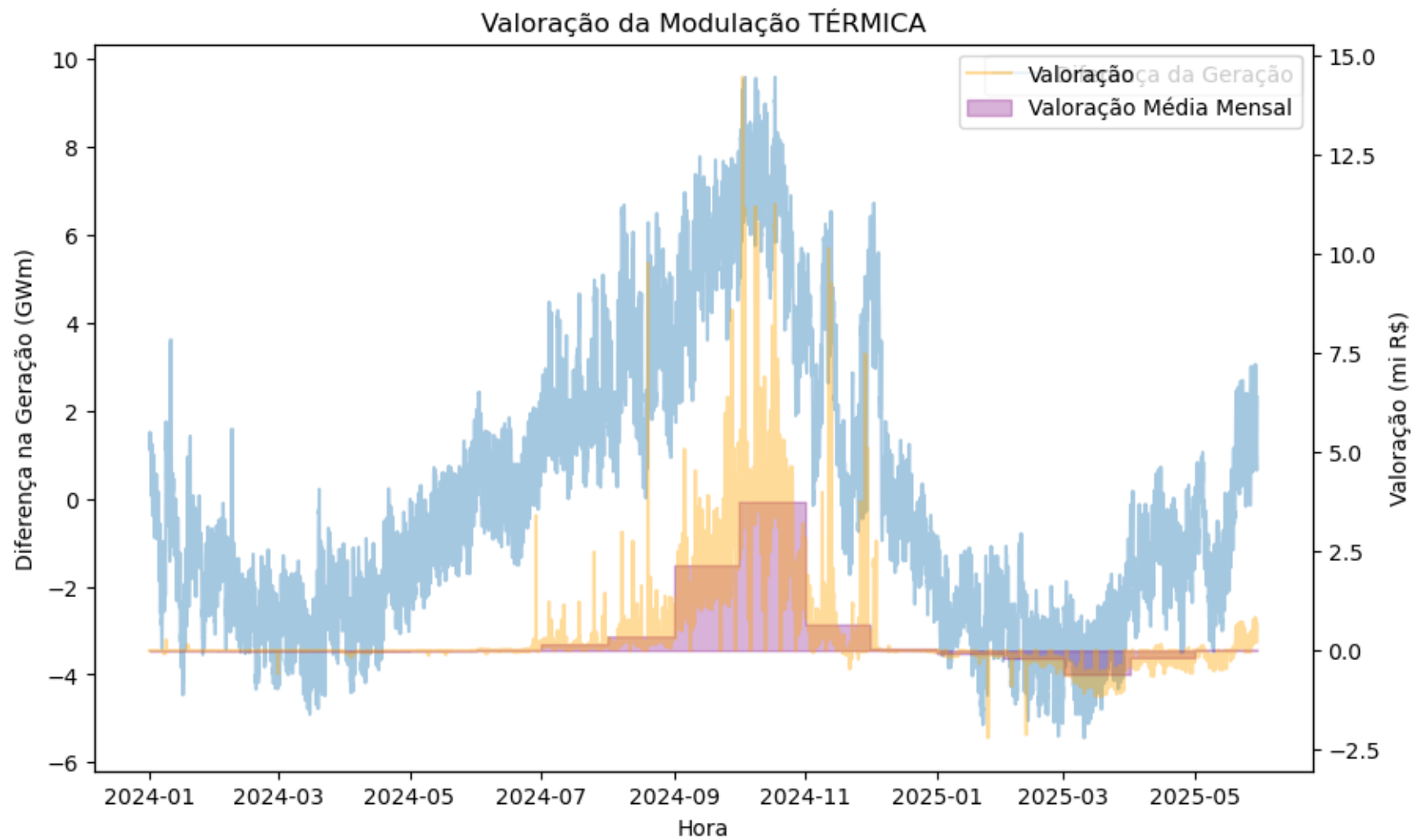


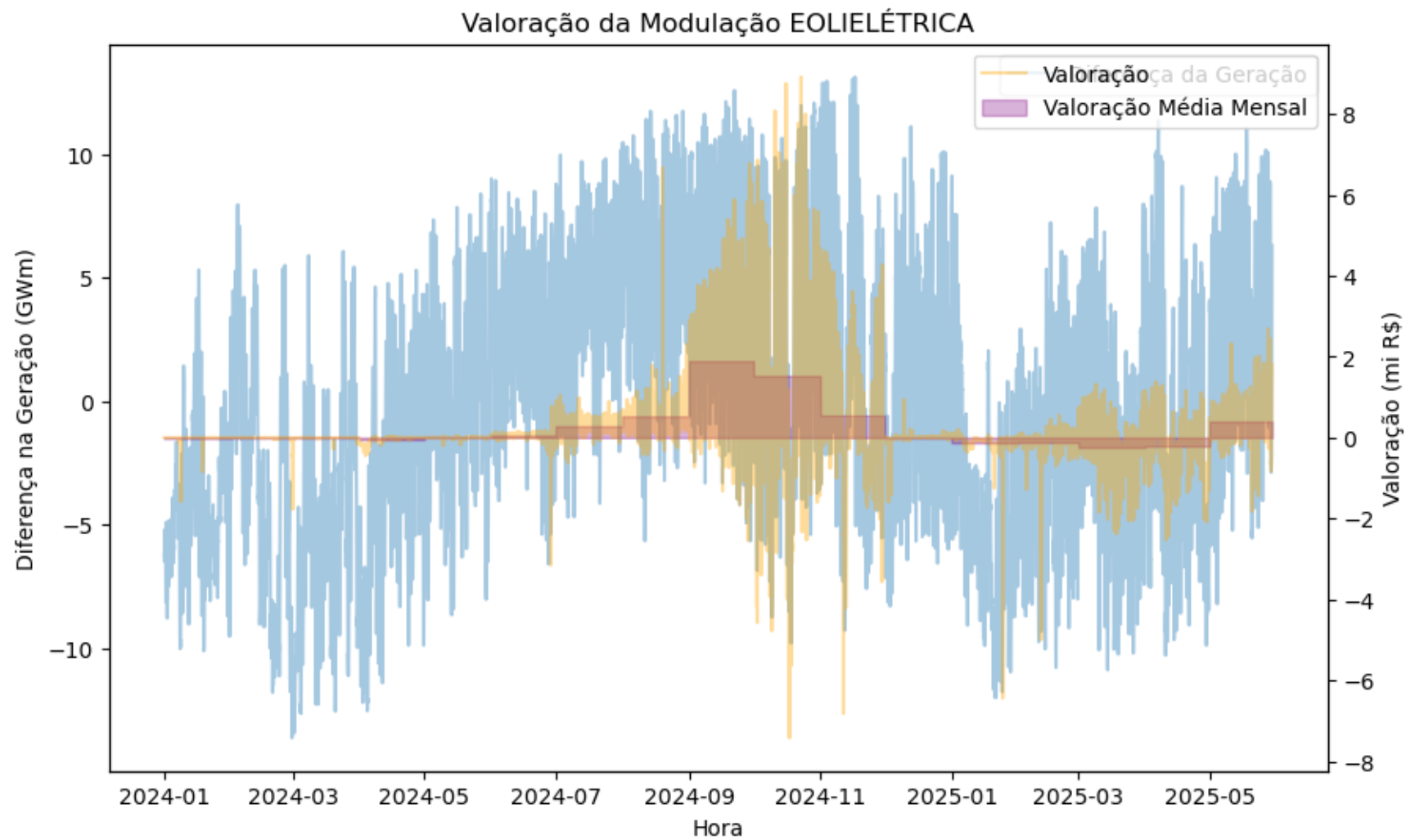


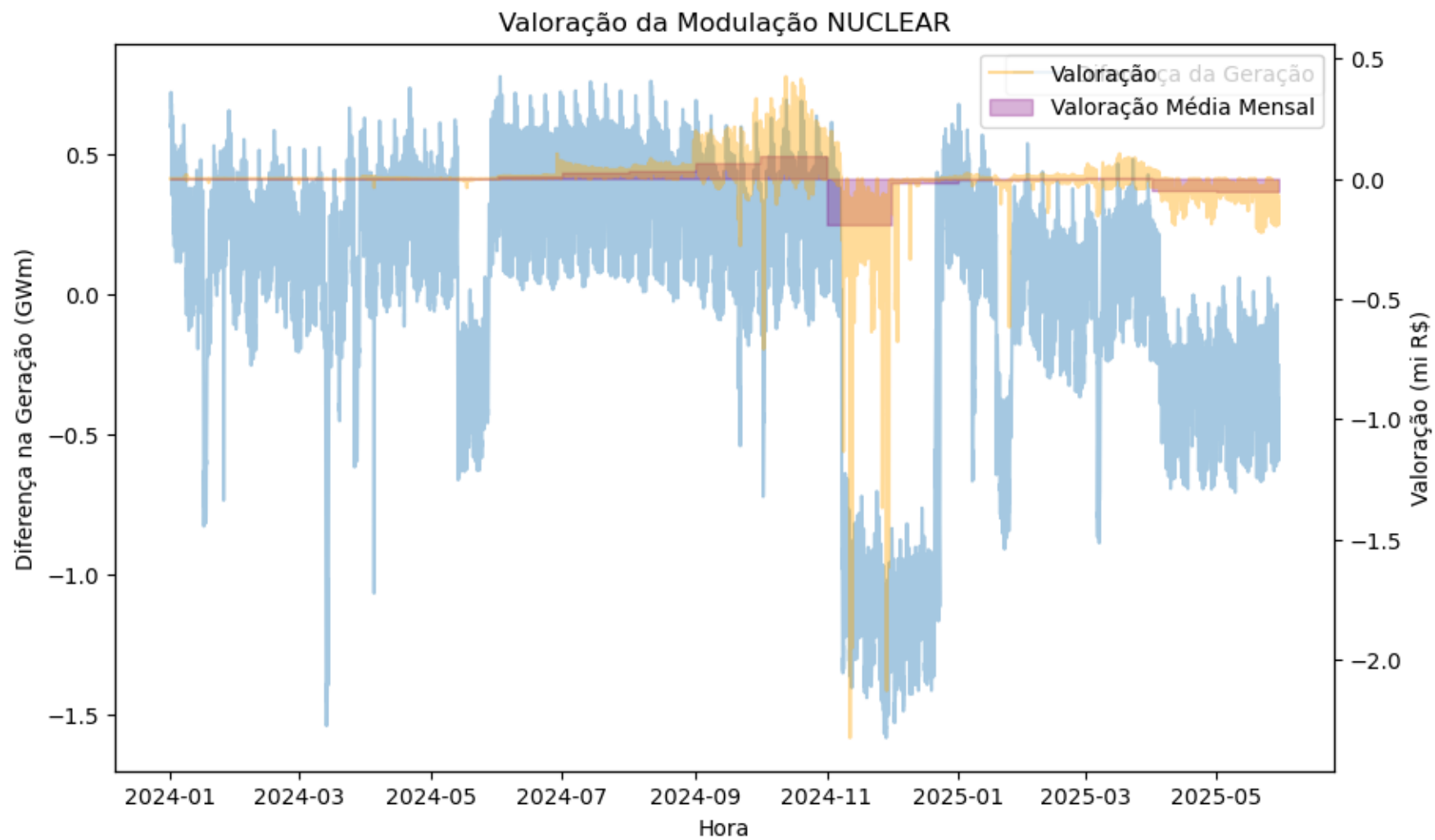














TÍTULO DA APRESENTAÇÃO

Subtítulo da apresentação